

第六章 局域网技术

The Medium Access Control Sublayer

主要内容

- 6.1 通道分配问题
- 6.2 多路访问协议
- 6.3 Ethernet
- 6.4 无线局域网Wireless LAN
- 6.5 宽带无线通信
- 6.6 桥接器、交换机和路由器

6.1 通道分配问题

- 局域网络的关键技术问题是：
 - 如何分配传输信道
- 静态通道分配： LANs and MANs
 - FDM
 - TDM
- 动态通道分配： LANs and MANs

局域和城域网络中的动态通道分配

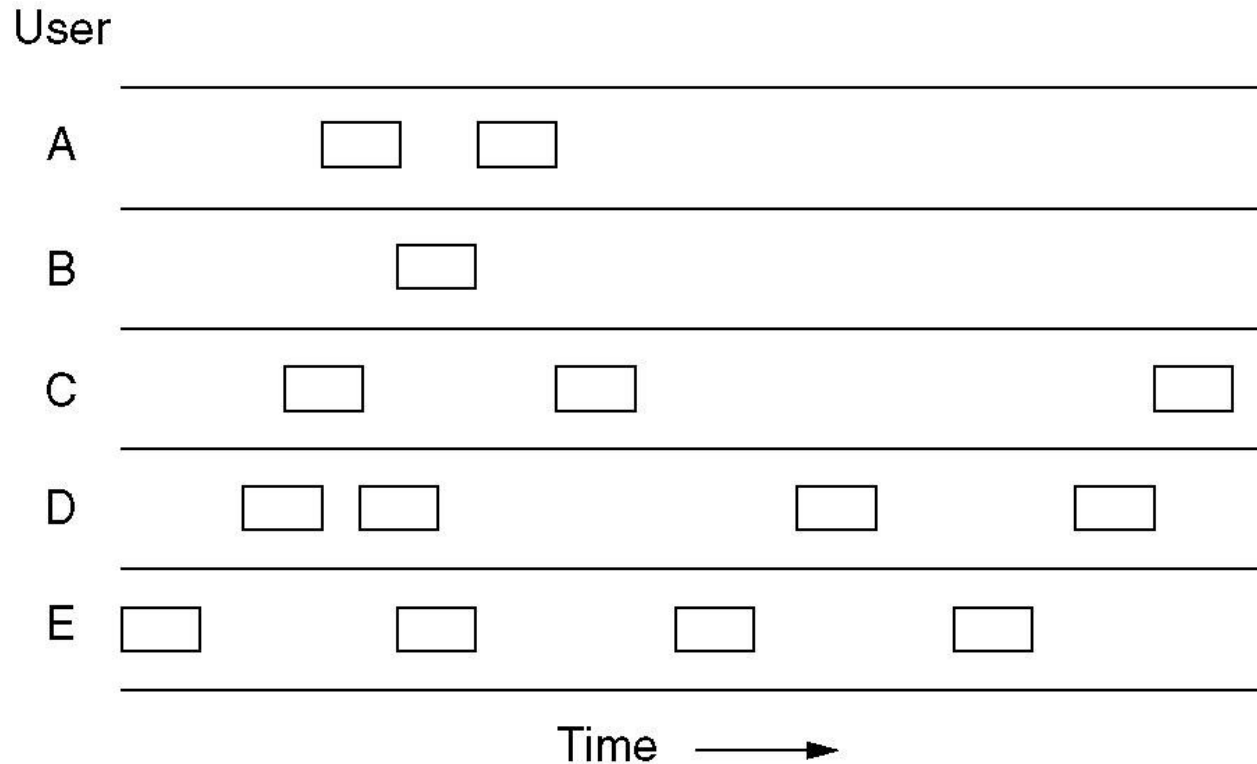
1. Station Model.
2. Single Channel Assumption. 单一通道
3. Collision Assumption. 碰撞
4. (a) Continuous Time. 连续时间
(b) Slotted Time. 分槽时间
5. (a) Carrier Sense. 载波监听
(b) No Carrier Sense. 非载波监听

6.2 多路访问协议

Multiple Access Protocols

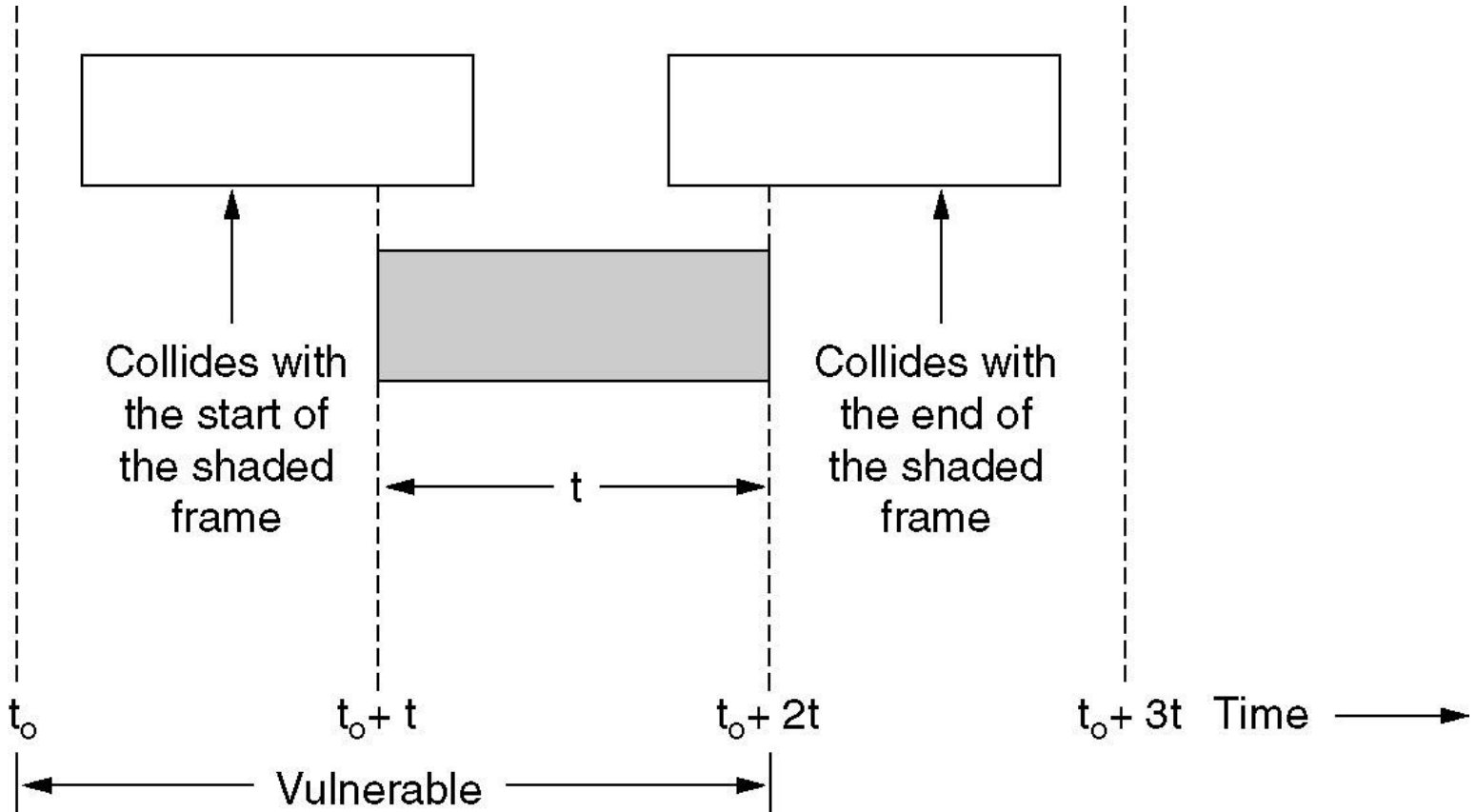
- ALOHA
- Carrier Sense Multiple Access Protocols
 - 载波监听多路访问协议
- Collision-Free Protocols无碰撞协议
- Limited-Contention Protocols有限争夺协议
- Wavelength Division Multiple Access Protocols 波分多路访问协议
- Wireless LAN Protocols无线局域网协议

Pure ALOHA



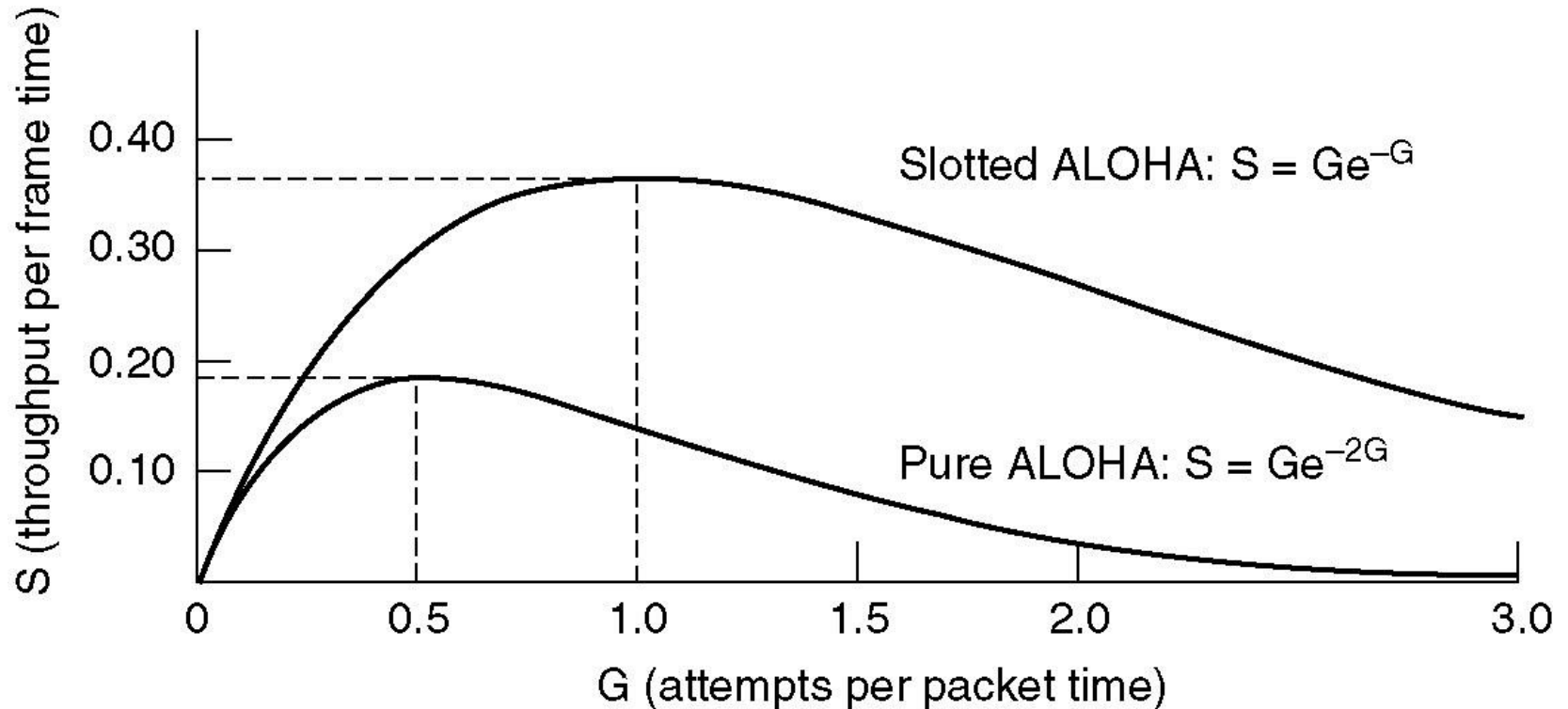
In pure ALOHA, frames are transmitted at completely arbitrary times.

Pure ALOHA (2)



Vulnerable period for the shaded frame.

Pure ALOHA (3)



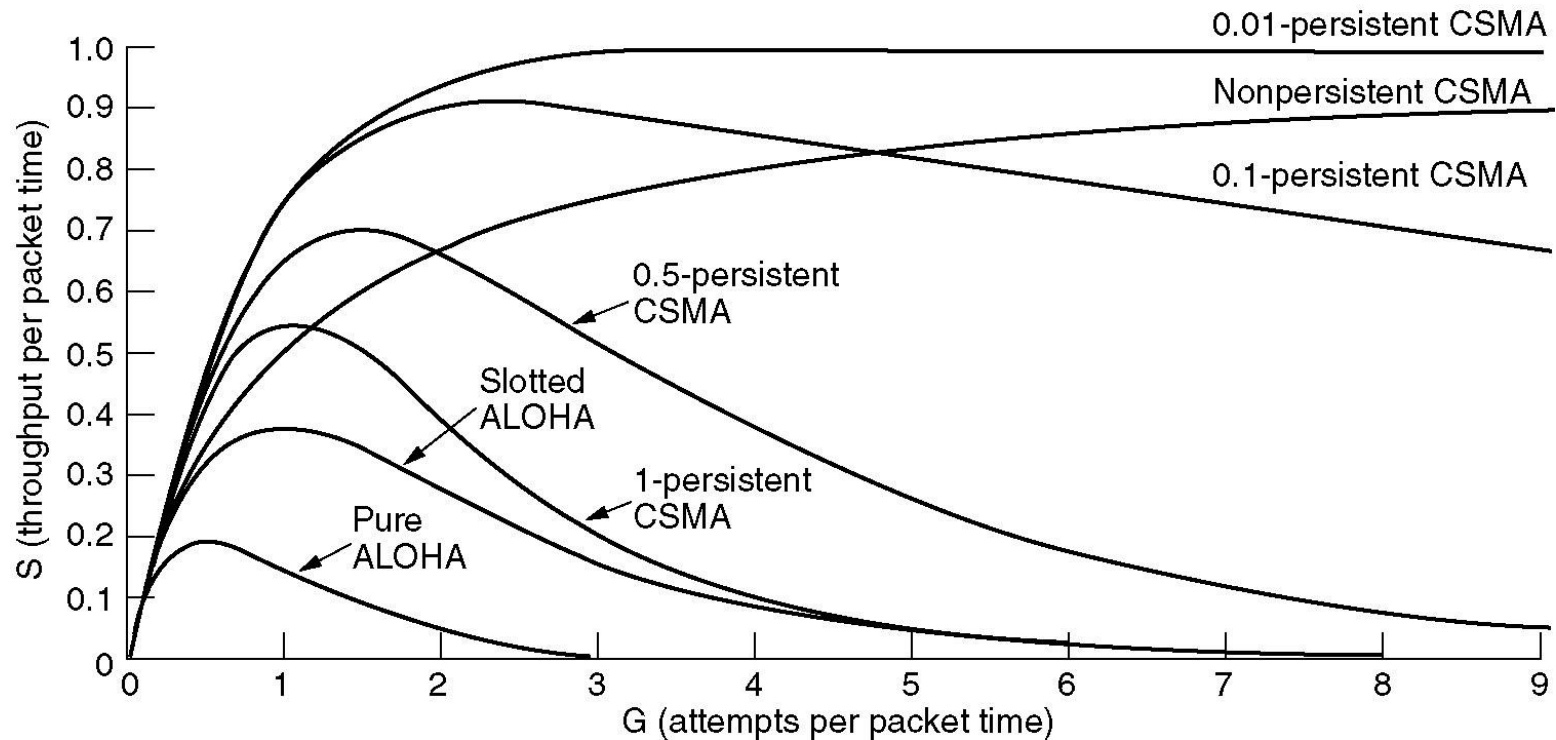
Throughput versus offered traffic for ALOHA systems.

载波监听多路访问CSMA

- CSMA 基本原理
- 1-坚持型 CSMA
 - 发送前监听，通道不空，继续监听，直到通道空，发送。优：通道利用率高；缺：冲突大。
- 非坚持型 CSMA
 - 发送前监听，通道空，发送；通道不空，不继续监听，等待一个概率时间，继续监听，直到通道空，发送。
 - 优：通道利用率高，冲突小；缺：延迟大。
- P-坚持型 CSMA（分时间槽通道）
 - 发送前在时间片槽前监听，通道空，以概率 p 发送；通道不空，继续监听，等待下一个时间片槽，继续监听。

坚持型和非坚持型的CSMA

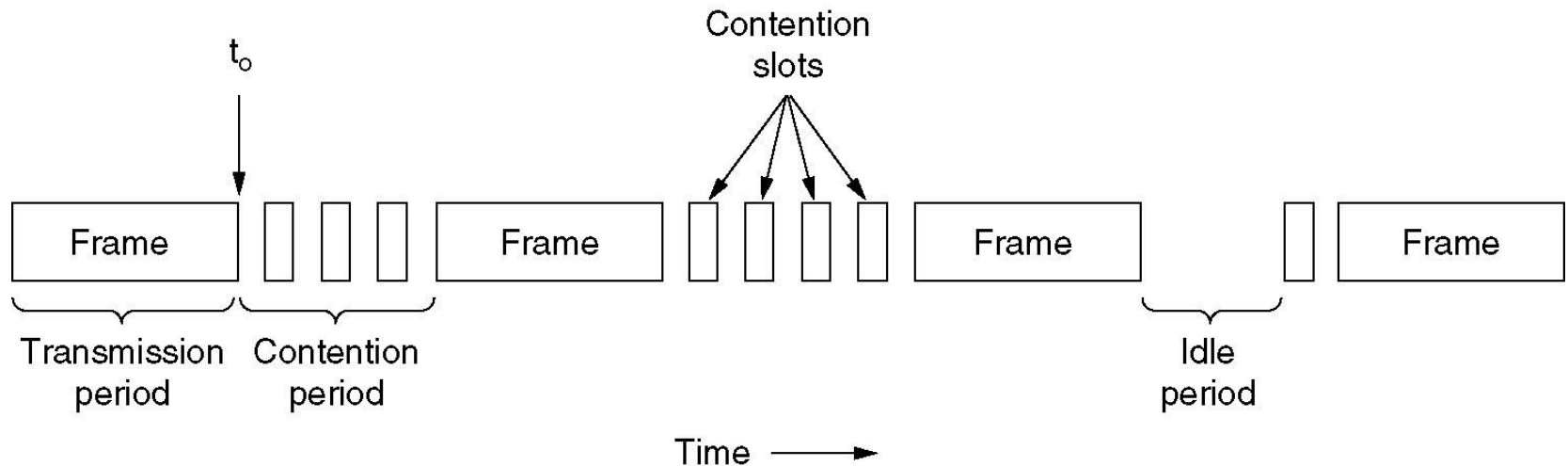
Persistent and Nonpersistent CSMA



Comparison of the channel utilization versus load for various random access protocols.

帶冲突检测的载波监听多路访问

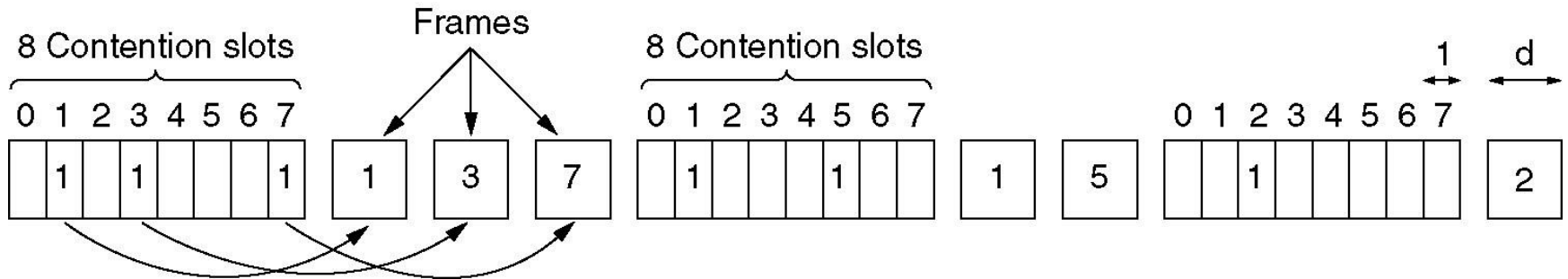
CSMA with Collision Detection



CSMA/CD can be in one of three states: contention, transmission, or idle.

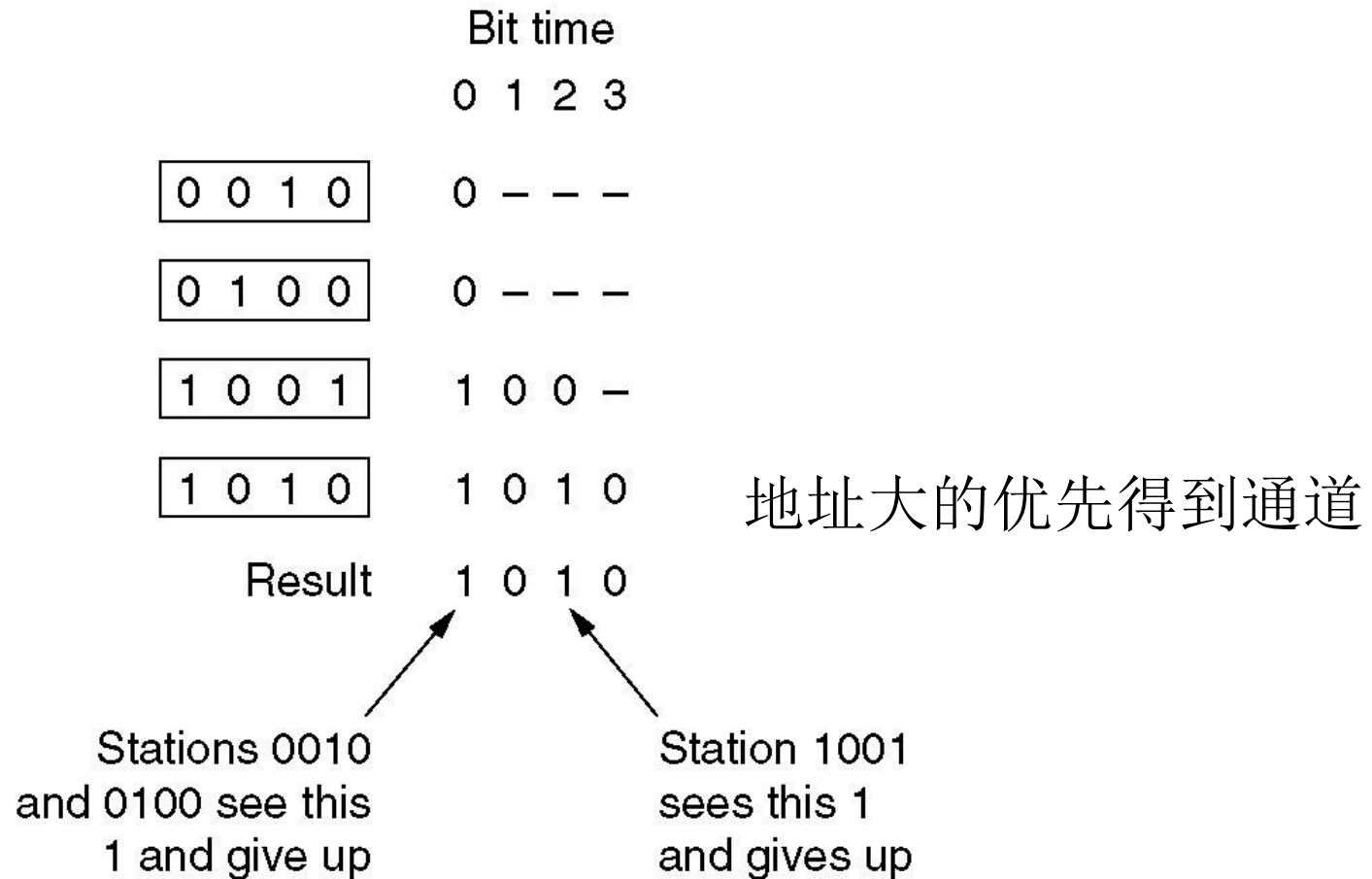
无冲突协议

Collision-Free Protocols



The basic bit-map protocol

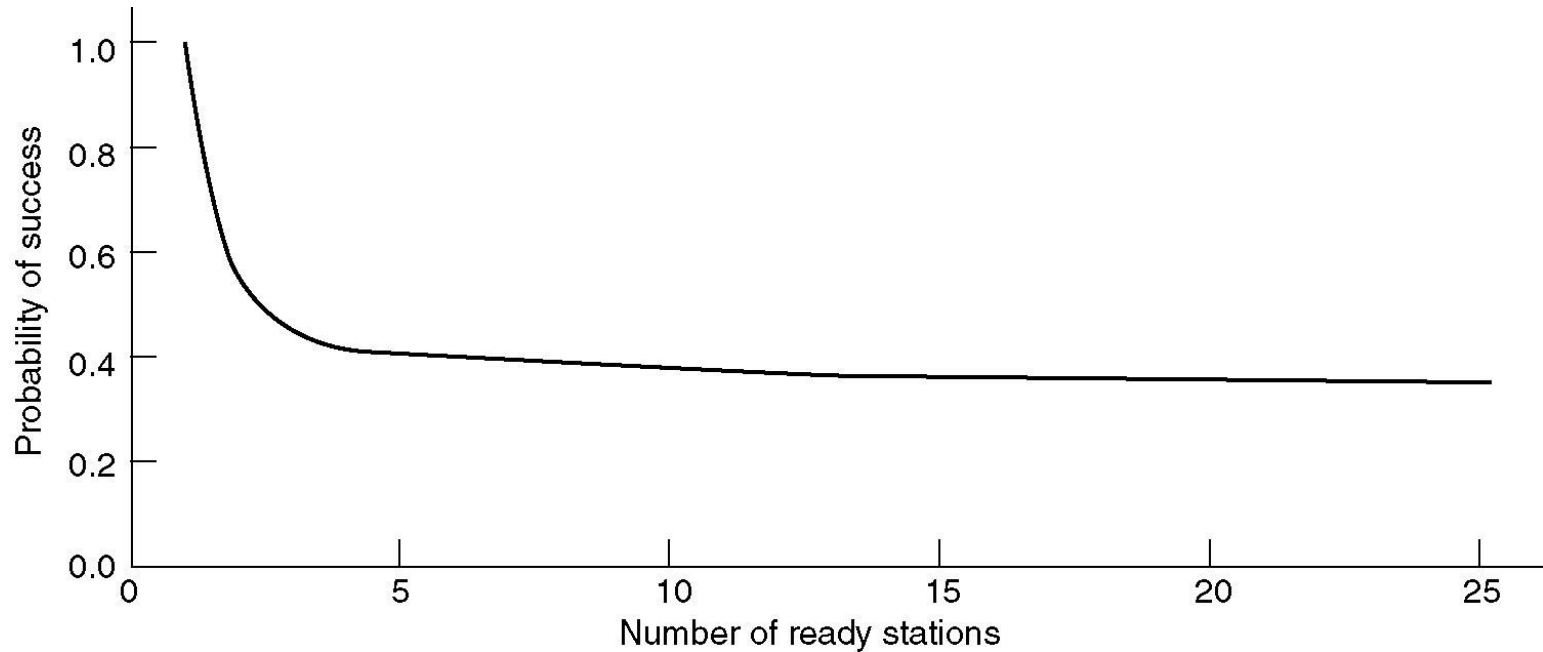
无冲突协议 Collision-Free Protocols



二进制倒数协议 The binary countdown protocol
A dash indicates silence

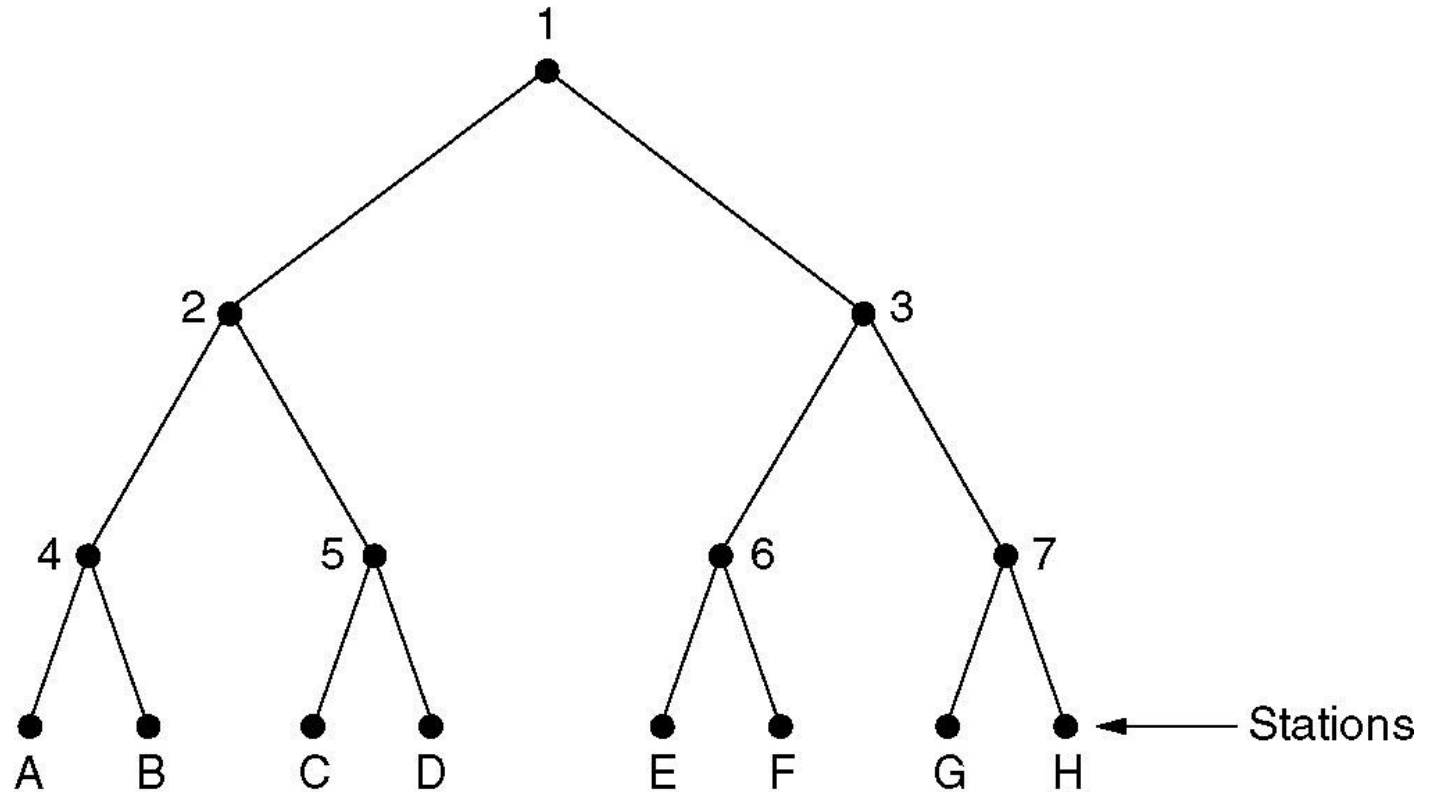
有限争抢协议

Limited-Contention Protocols



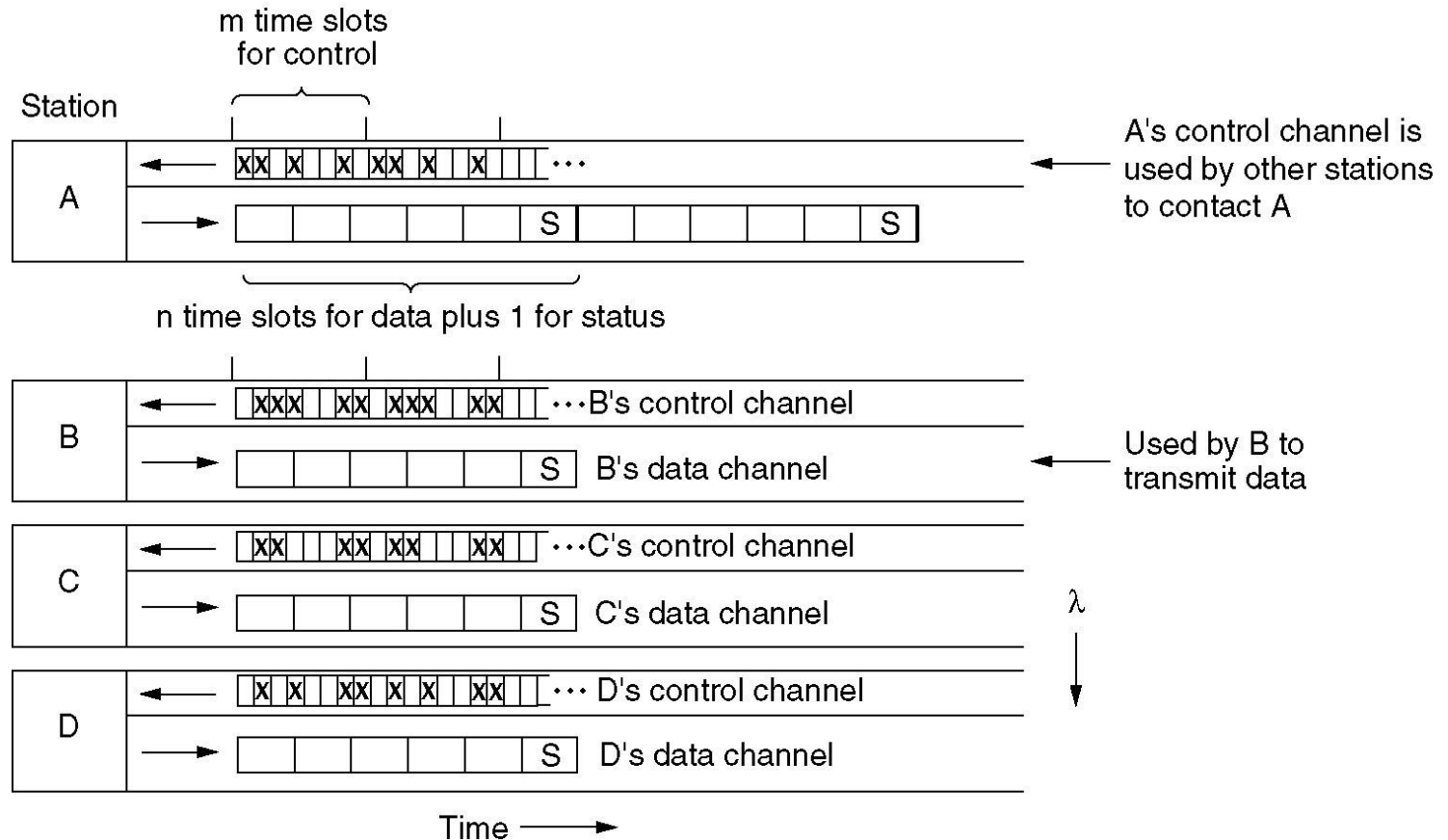
Acquisition probability for a symmetric contention channel.

Adaptive Tree Walk Protocol



The tree for eight stations.

Wavelength Division Multiple Access Protocols



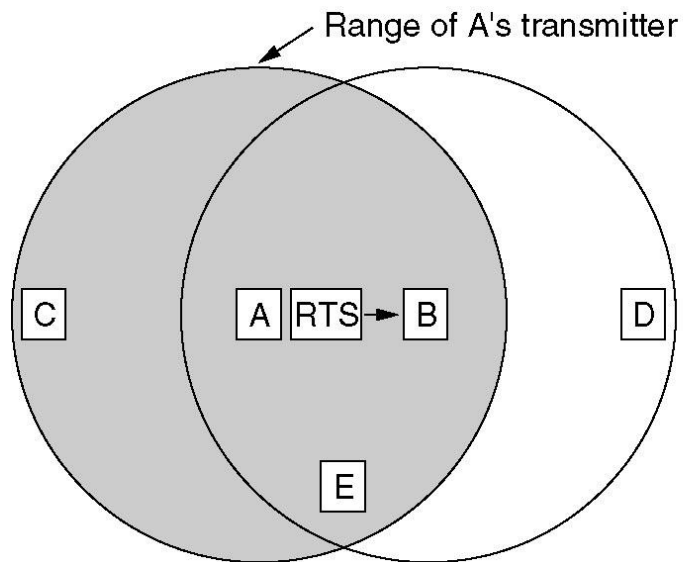
Wavelength division multiple access.

Wireless LAN Protocols

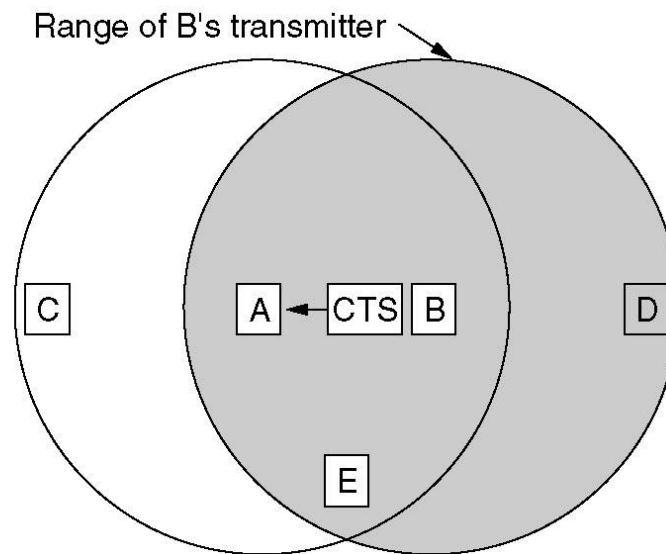


A wireless LAN. (a) A transmitting. (b) B transmitting.

Wireless LAN Protocols (2)



(a)



(b)

The MACA protocol. 避免冲突的多路访问协议

(a) A sending an RTS to B. (b) B responding with a CTS to A.

6.3 局域网和城域网标准

- IEEE 802.2 LLC 协议
- IEEE 802.3 和 Ethernet
- IEEE 802.4 和 Token Bus
- IEEE 802.5 和 Token Ring
- IEEE 802.11 Wireless LAN 无线局域网协议
- IEEE 802.16 无线城域网协议

Ethernet

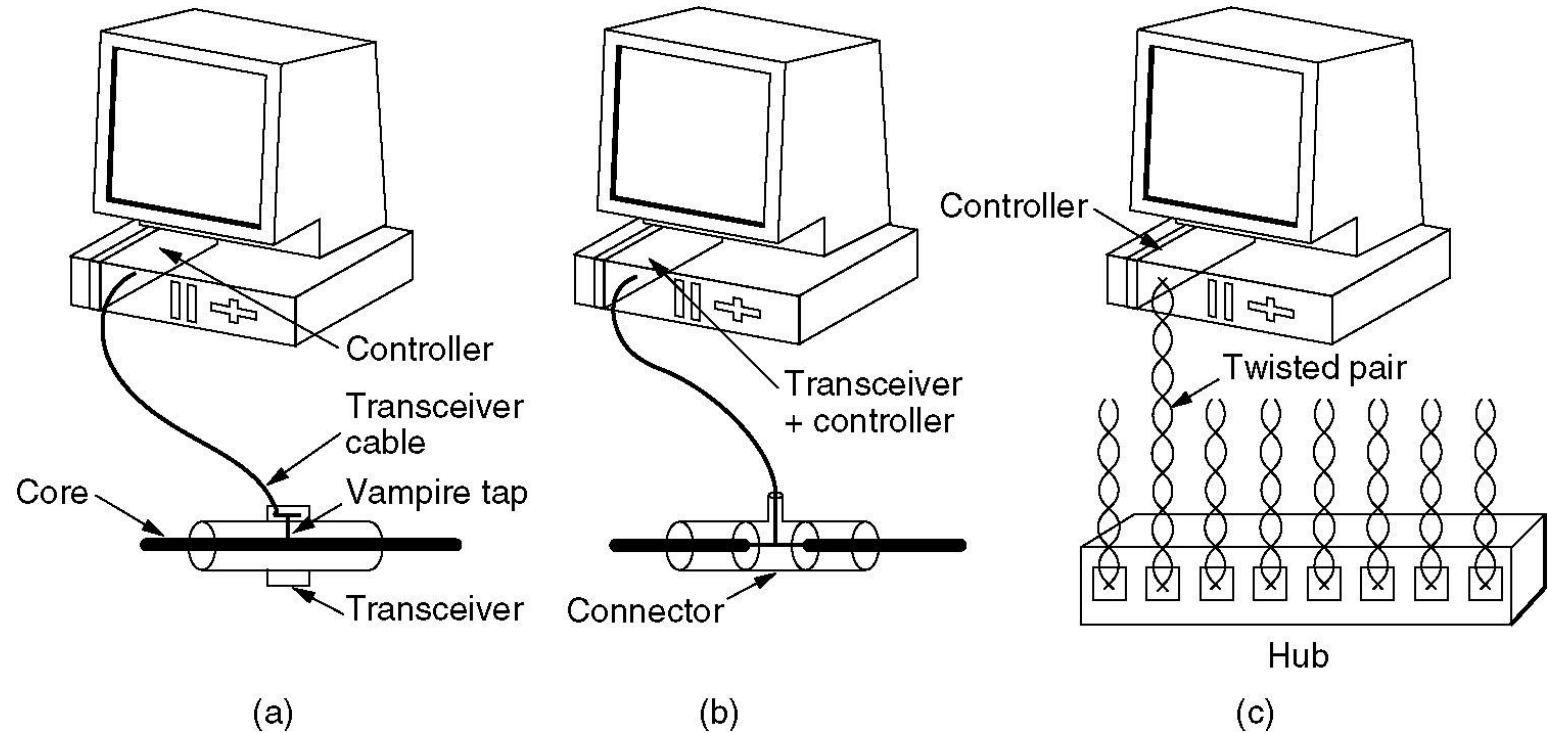
- Ethernet Cabling
- Manchester Encoding
- The Ethernet MAC Sublayer Protocol
- The Binary Exponential Backoff Algorithm
- Ethernet Performance
- Switched Ethernet
- Fast Ethernet
- Gigabit Ethernet
- IEEE 802.2: Logical Link Control
- Retrospective on Ethernet

Ethernet Cabling

Name	Cable	Max. seg.	Nodes/seg.	Advantages
10Base5	Thick coax	500 m	100	Original cable; now obsolete
10Base2	Thin coax	185 m	30	No hub needed
10Base-T	Twisted pair	100 m	1024	Cheapest system
10Base-F	Fiber optics	2000 m	1024	Best between buildings

The most common kinds of Ethernet cabling.

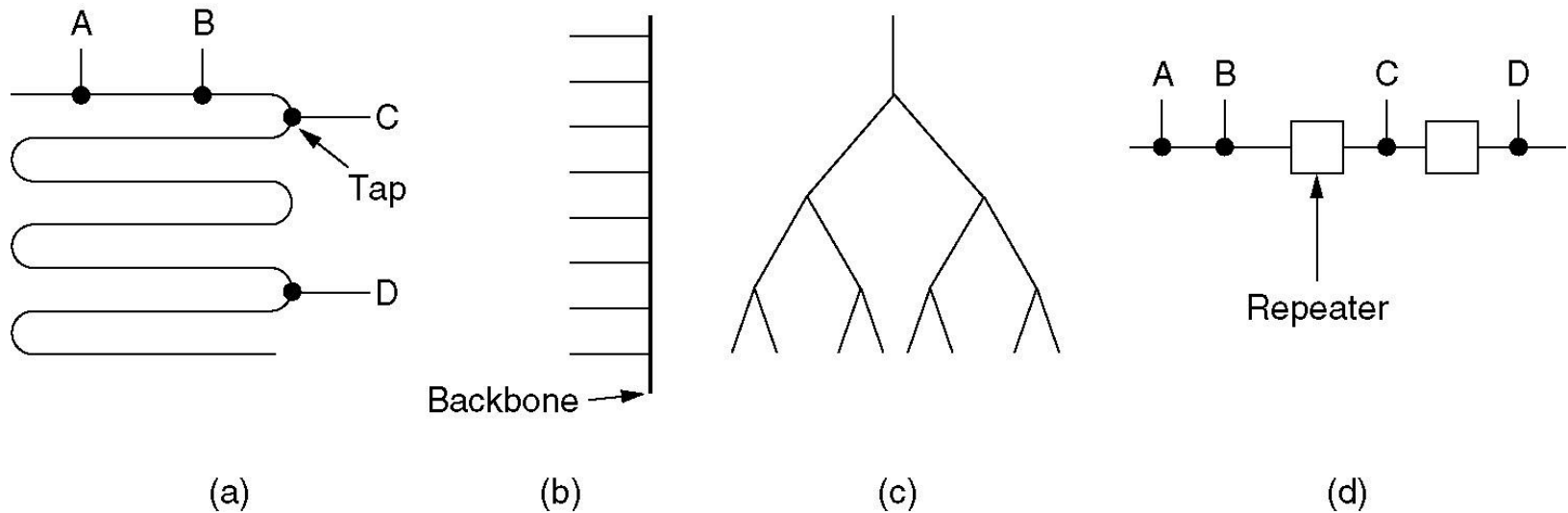
Ethernet Cabling (2)



Three kinds of Ethernet cabling.

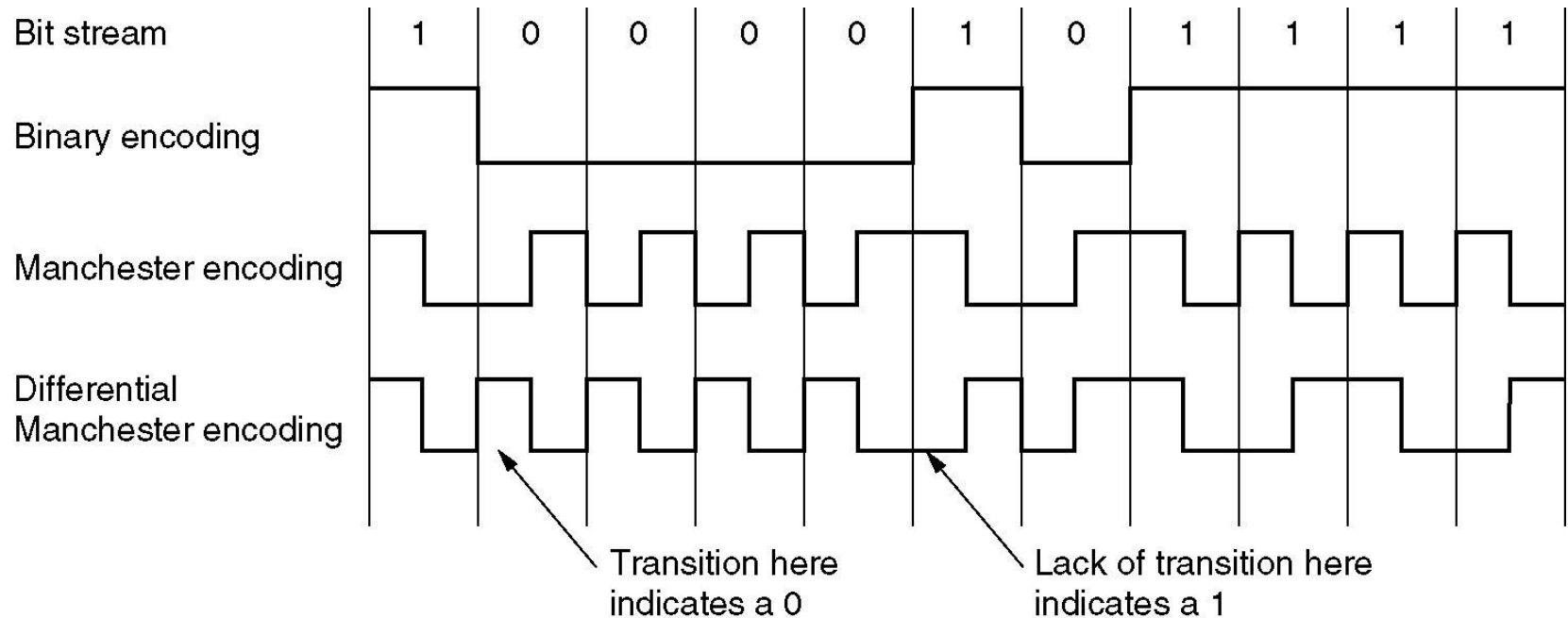
(a) 10Base5, (b) 10Base2, (c) 10Base-T.

Ethernet Cabling (3)



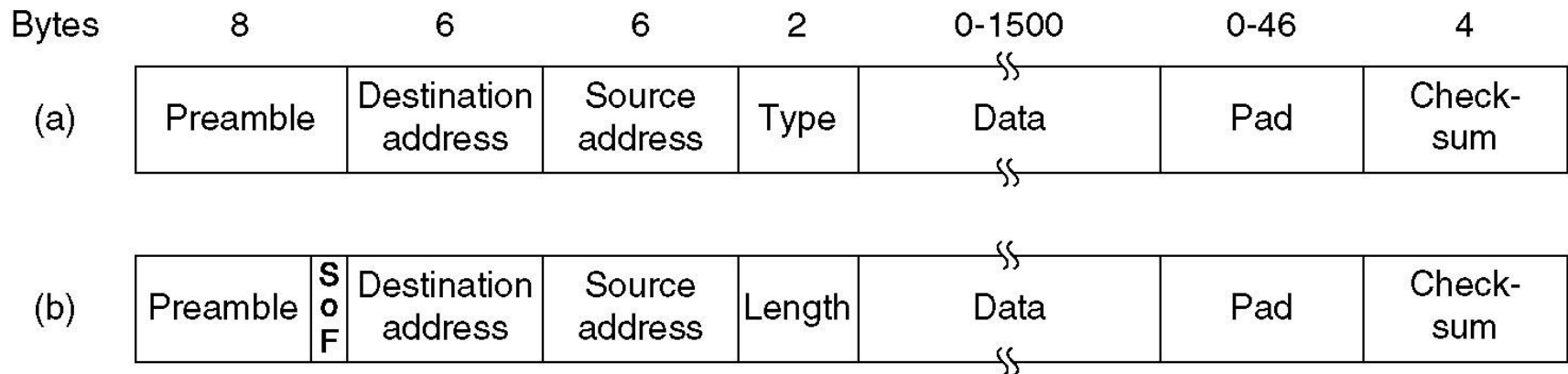
Cable topologies. (a) Linear, (b) Spine,
(c) Tree, (d) Segmented.

Ethernet Cabling (4)



(a) Binary encoding, (b) Manchester encoding,
(c) Differential Manchester encoding.

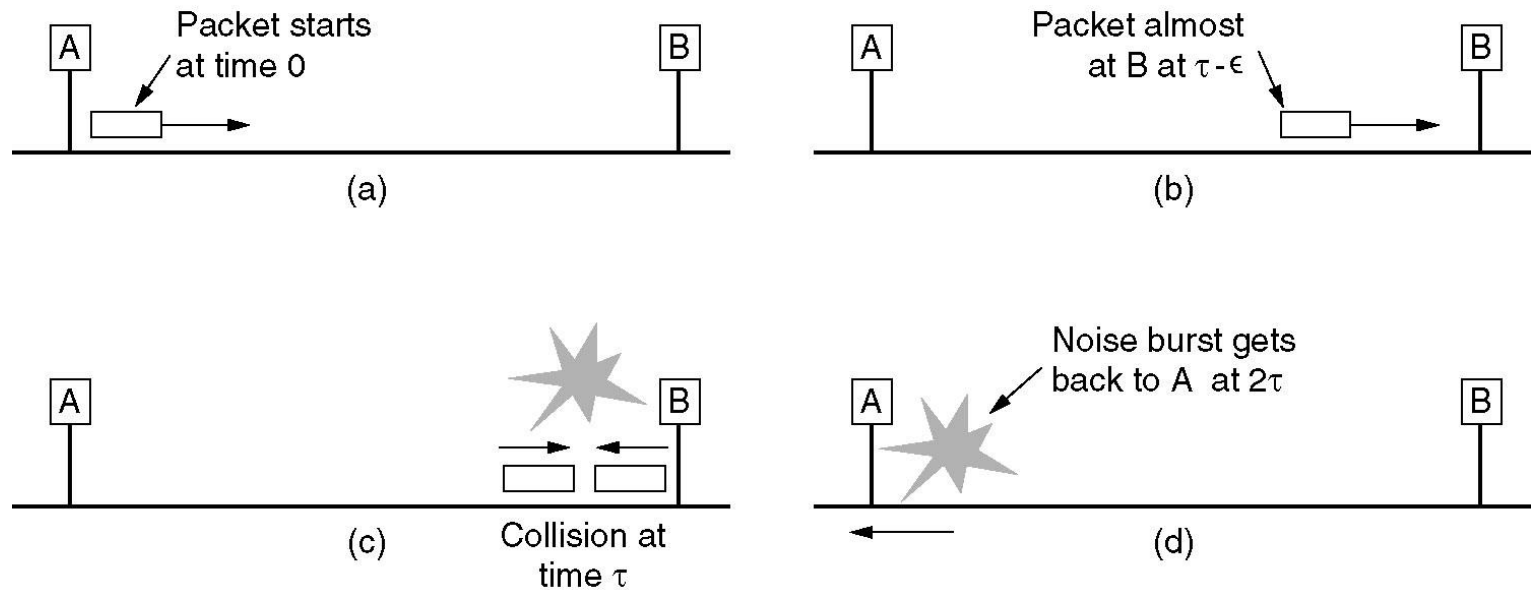
Ethernet MAC Sublayer Protocol



Frame formats. (a) DIX Ethernet, (b) IEEE 802.3.

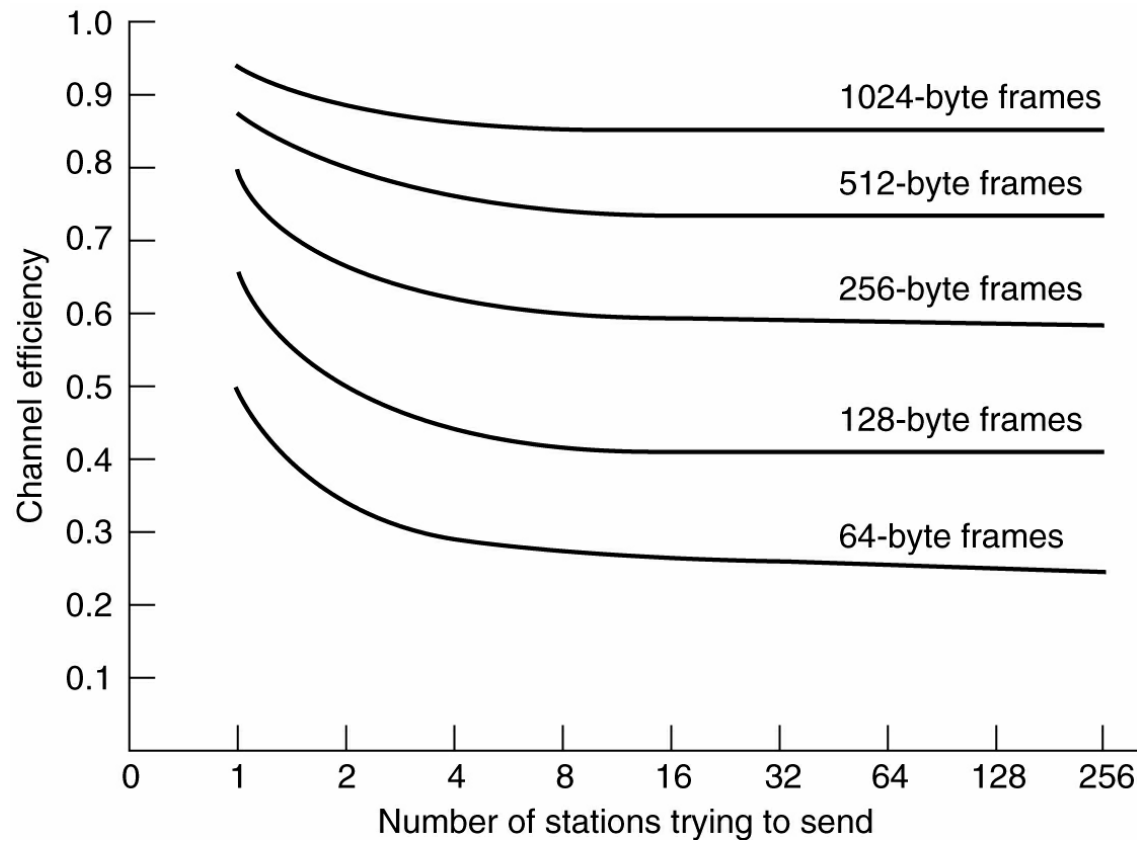
Ethernet MAC Sublayer Protocol (2)

二进制指数后退协议



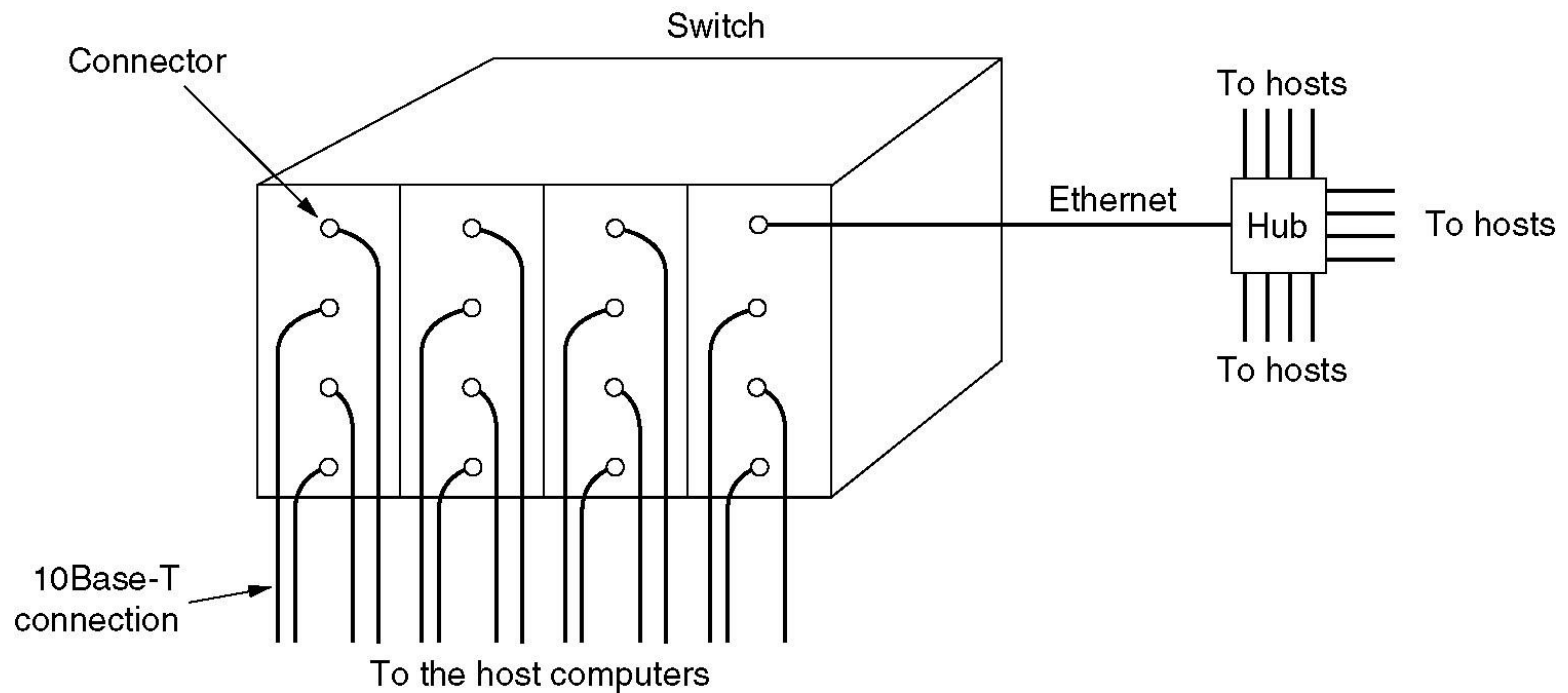
Collision detection can take as long as $2t$

Ethernet Performance



Efficiency of Ethernet at 10 Mbps with 512-bit slot times.

Switched Ethernet



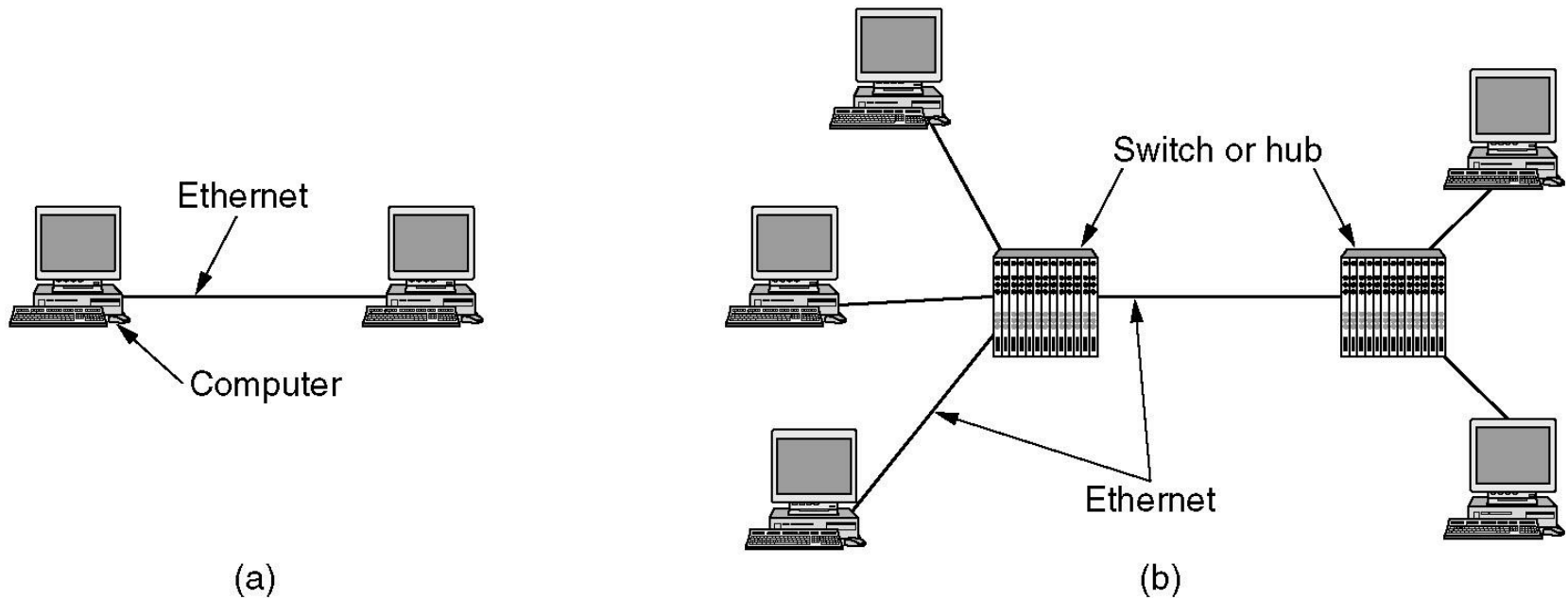
A simple example of switched Ethernet.

Fast Ethernet

Name	Cable	Max. segment	Advantages
100Base-T4	Twisted pair	100 m	Uses category 3 UTP
100Base-TX	Twisted pair	100 m	Full duplex at 100 Mbps
100Base-FX	Fiber optics	2000 m	Full duplex at 100 Mbps; long runs

The original fast Ethernet cabling.

Gigabit Ethernet



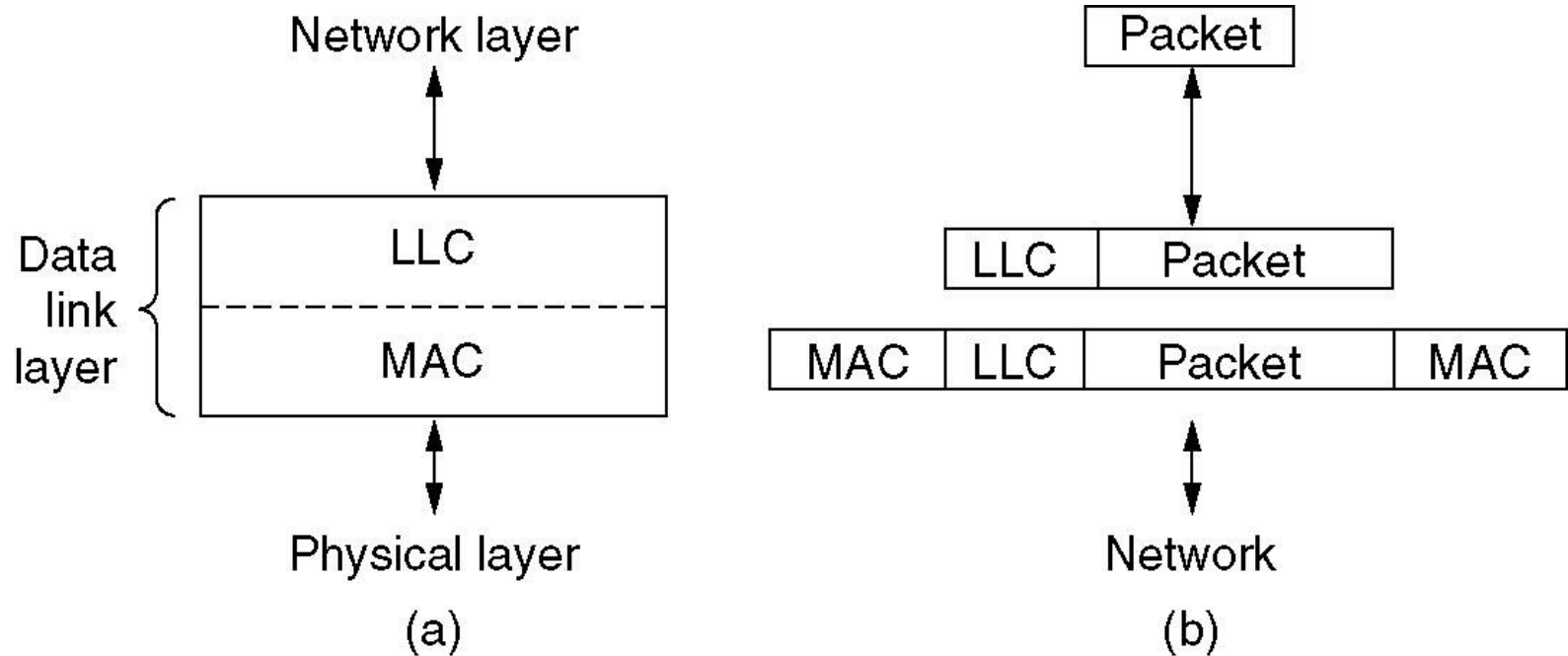
(a) A two-station Ethernet. **(b)** A multistation Ethernet.

Gigabit Ethernet (2)

Name	Cable	Max. segment	Advantages
1000Base-SX	Fiber optics	550 m	Multimode fiber (50, 62.5 microns)
1000Base-LX	Fiber optics	5000 m	Single (10 μ) or multimode (50, 62.5 μ)
1000Base-CX	2 Pairs of STP	25 m	Shielded twisted pair
1000Base-T	4 Pairs of UTP	100 m	Standard category 5 UTP

Gigabit Ethernet cabling.

IEEE 802.2: Logical Link Control

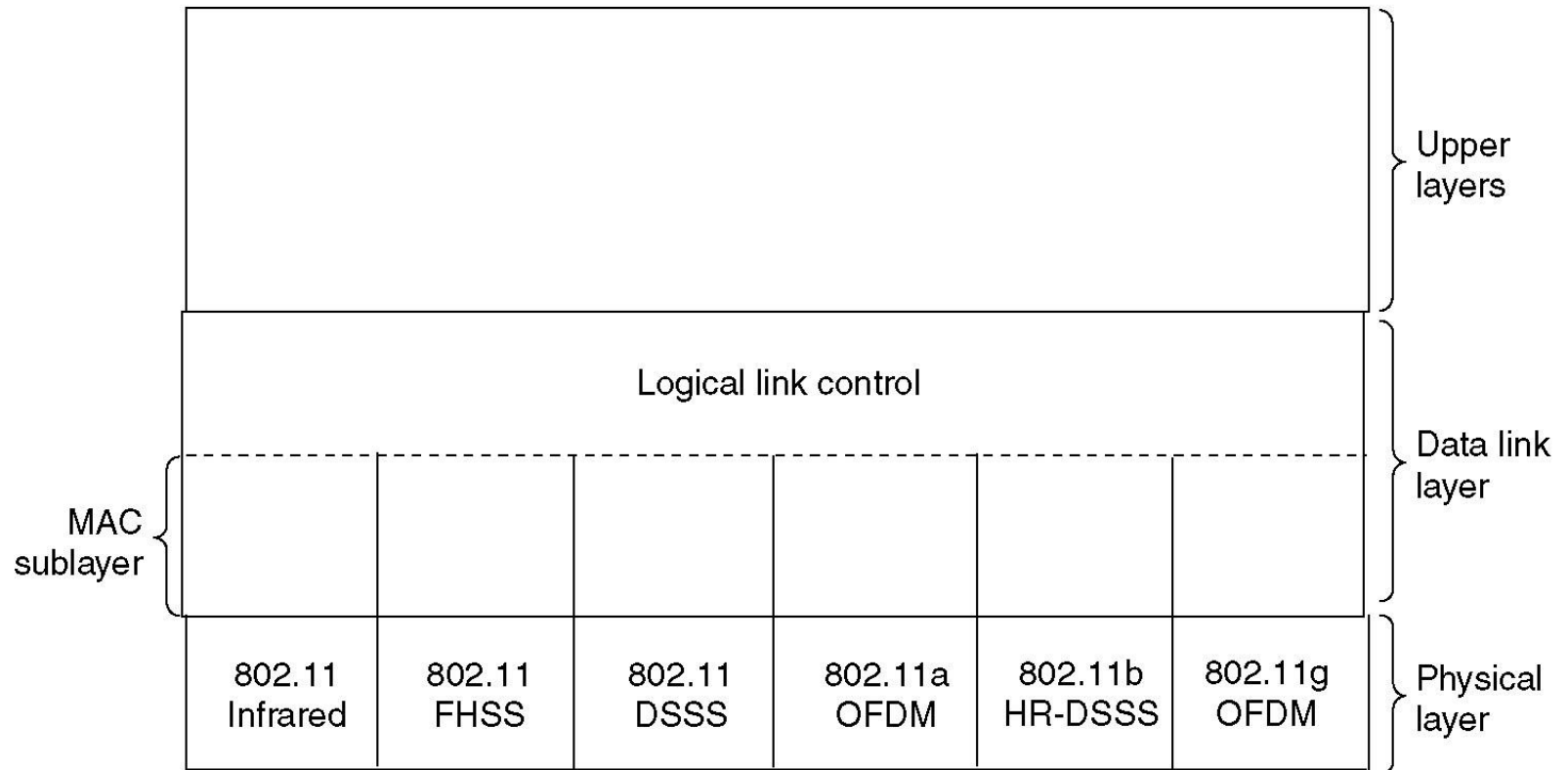


(a) Position of LLC. (b) Protocol formats.

6.4 Wireless LANs

- The 802.11 Protocol Stack
- The 802.11 Physical Layer
- The 802.11 MAC Sublayer Protocol
- The 802.11 Frame Structure
- Services

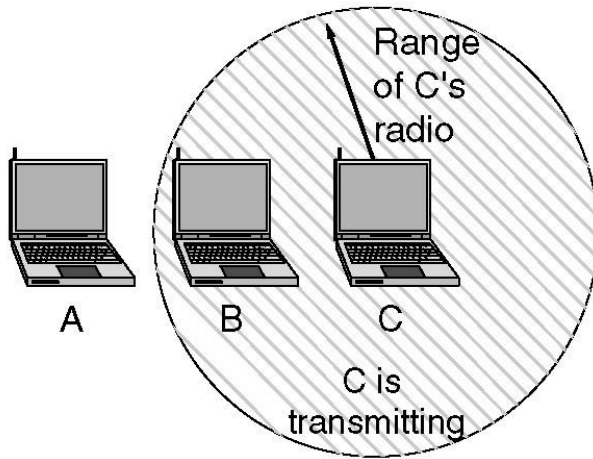
The 802.11 Protocol Stack



Part of the 802.11 protocol stack.

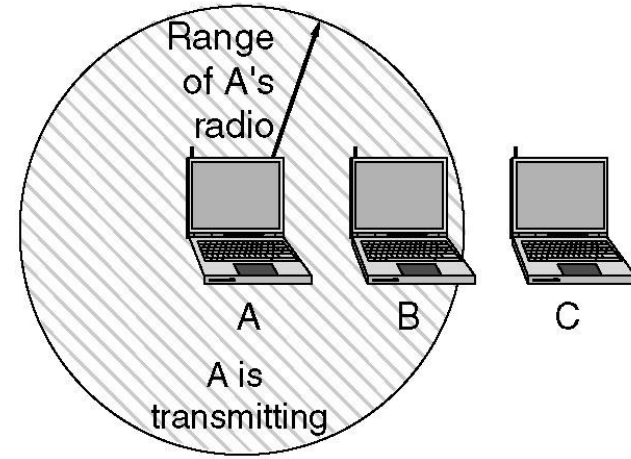
The 802.11 MAC Sublayer Protocol

A wants to send to B
but cannot hear that
B is busy



(a)

B wants to send to C
but mistakenly thinks
the transmission will fail

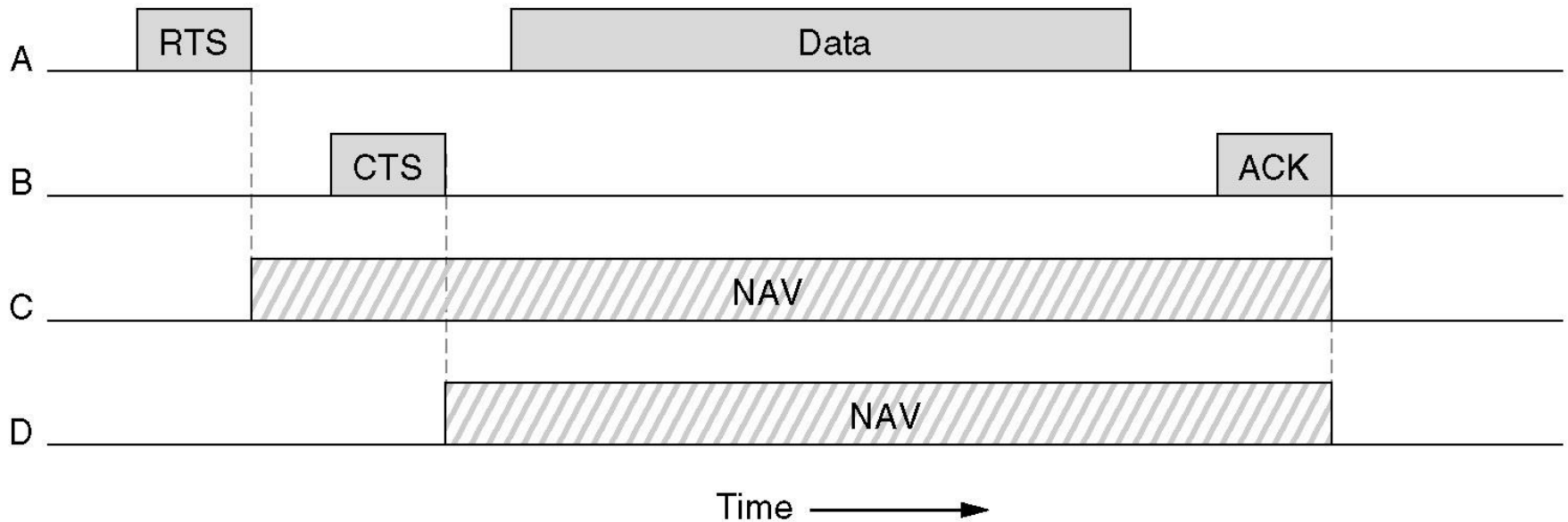


(b)

(a) The hidden station problem.

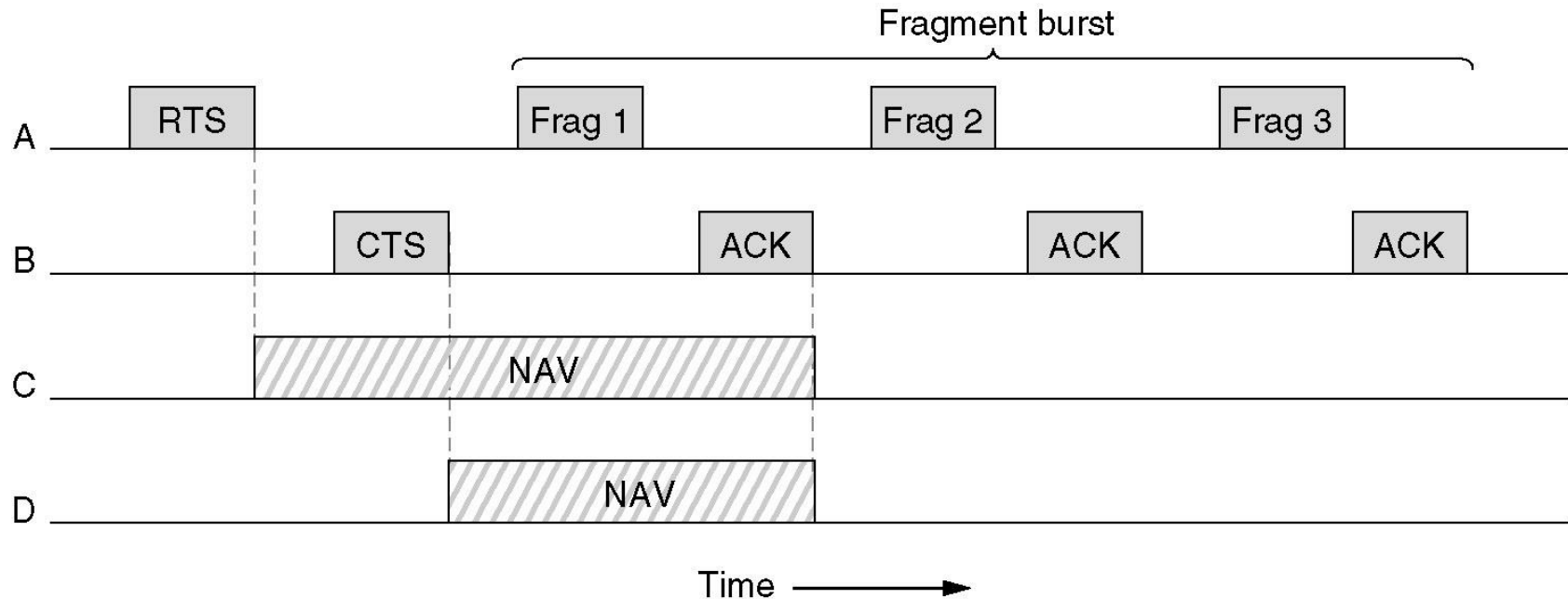
(b) The exposed station problem.

The 802.11 MAC Sublayer Protocol (2)



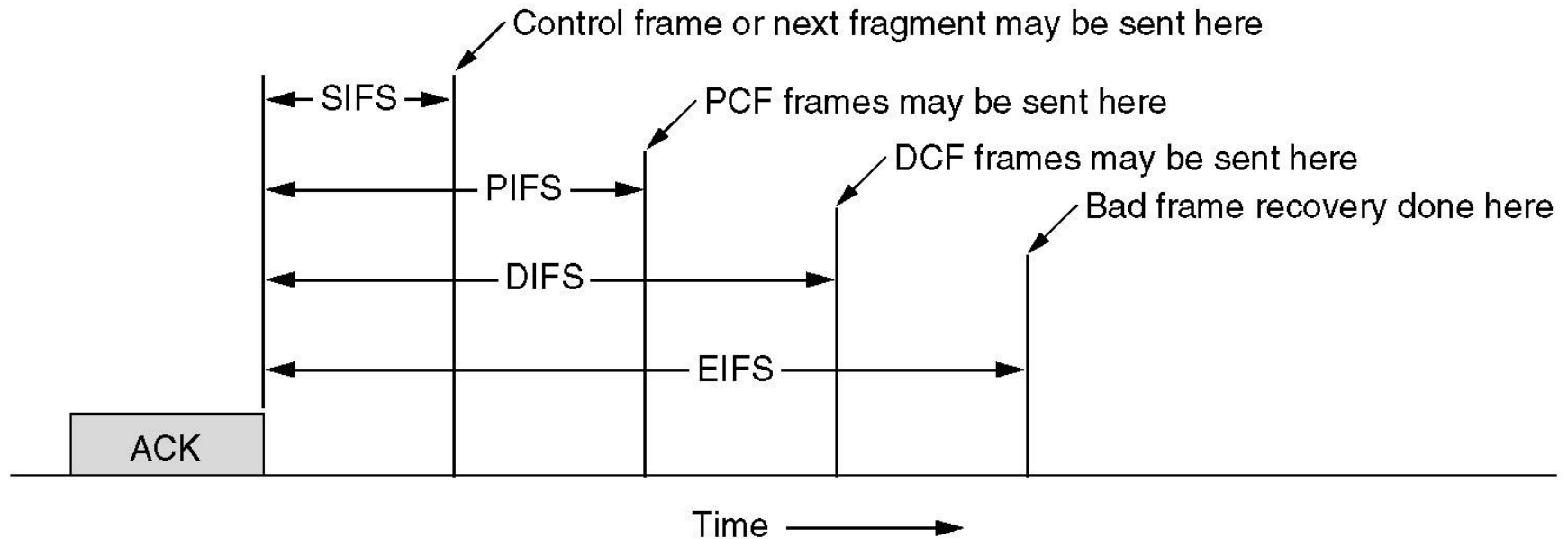
The use of virtual channel sensing using CSMA/CA.

The 802.11 MAC Sublayer Protocol (3)



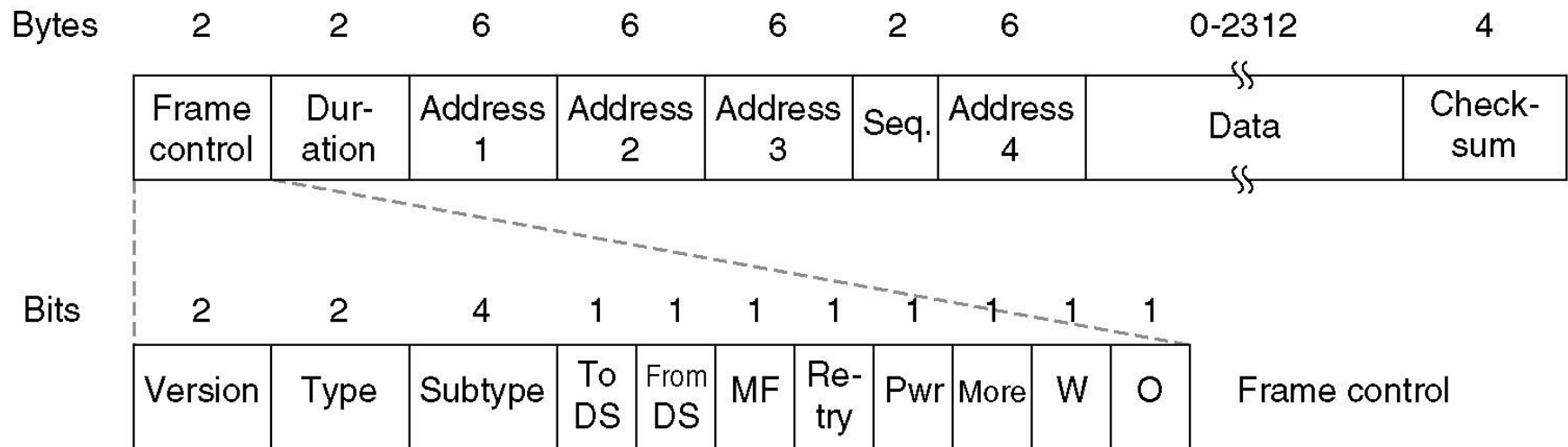
A fragment burst.

The 802.11 MAC Sublayer Protocol (4)



Interframe spacing in 802.11.

The 802.11 Frame Structure



The 802.11 data frame.

802.11 Services

Distribution Services

- Association
- Disassociation
- Reassociation
- Distribution
- Integration

802.11 Services

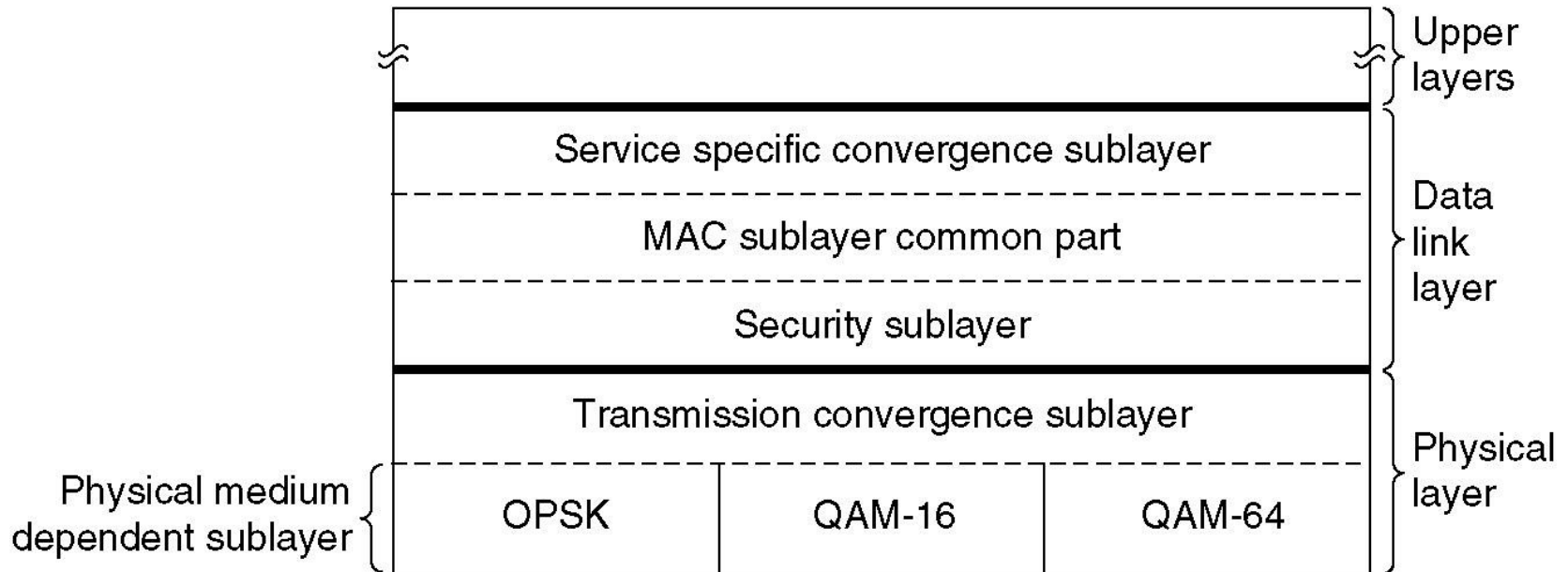
Intracell Services

- Authentication
- Deauthentication
- Privacy
- Data Delivery

6.5 Broadband Wireless

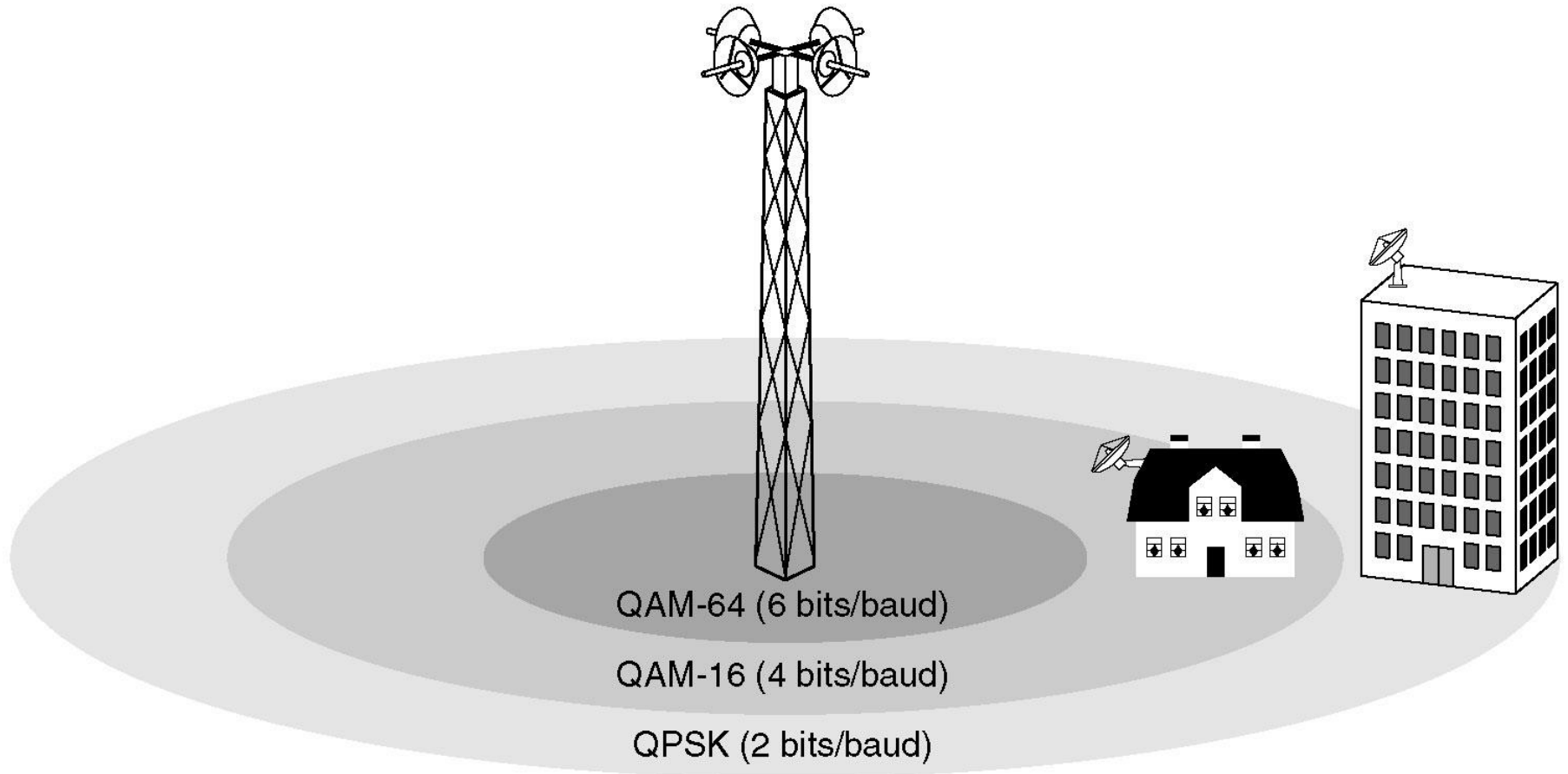
- Comparison of 802.11 and 802.16
- The 802.16 Protocol Stack
- The 802.16 Physical Layer
- The 802.16 MAC Sublayer Protocol
- The 802.16 Frame Structure

The 802.16 Protocol Stack



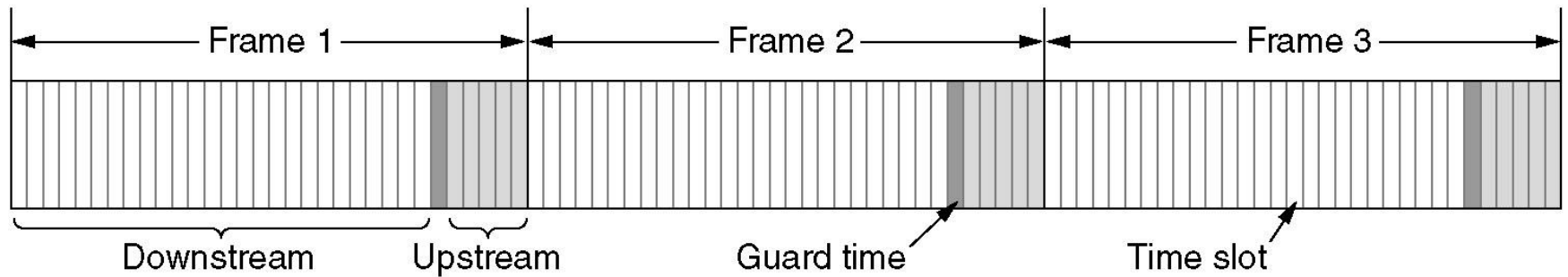
The 802.16 Protocol Stack.

The 802.16 Physical Layer



The 802.16 transmission environment.

The 802.16 Physical Layer (2)



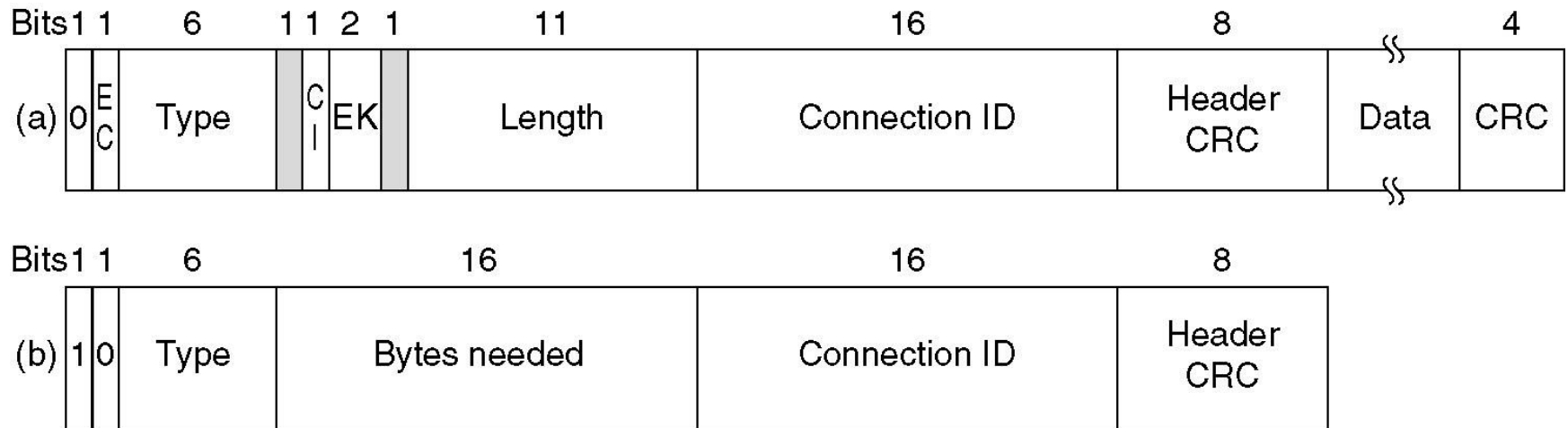
Frames and time slots for time division duplexing.

The 802.16 MAC Sublayer Protocol

Service Classes

- Constant bit rate service
- Real-time variable bit rate service
- Non-real-time variable bit rate service
- Best efforts service

The 802.16 Frame Structure

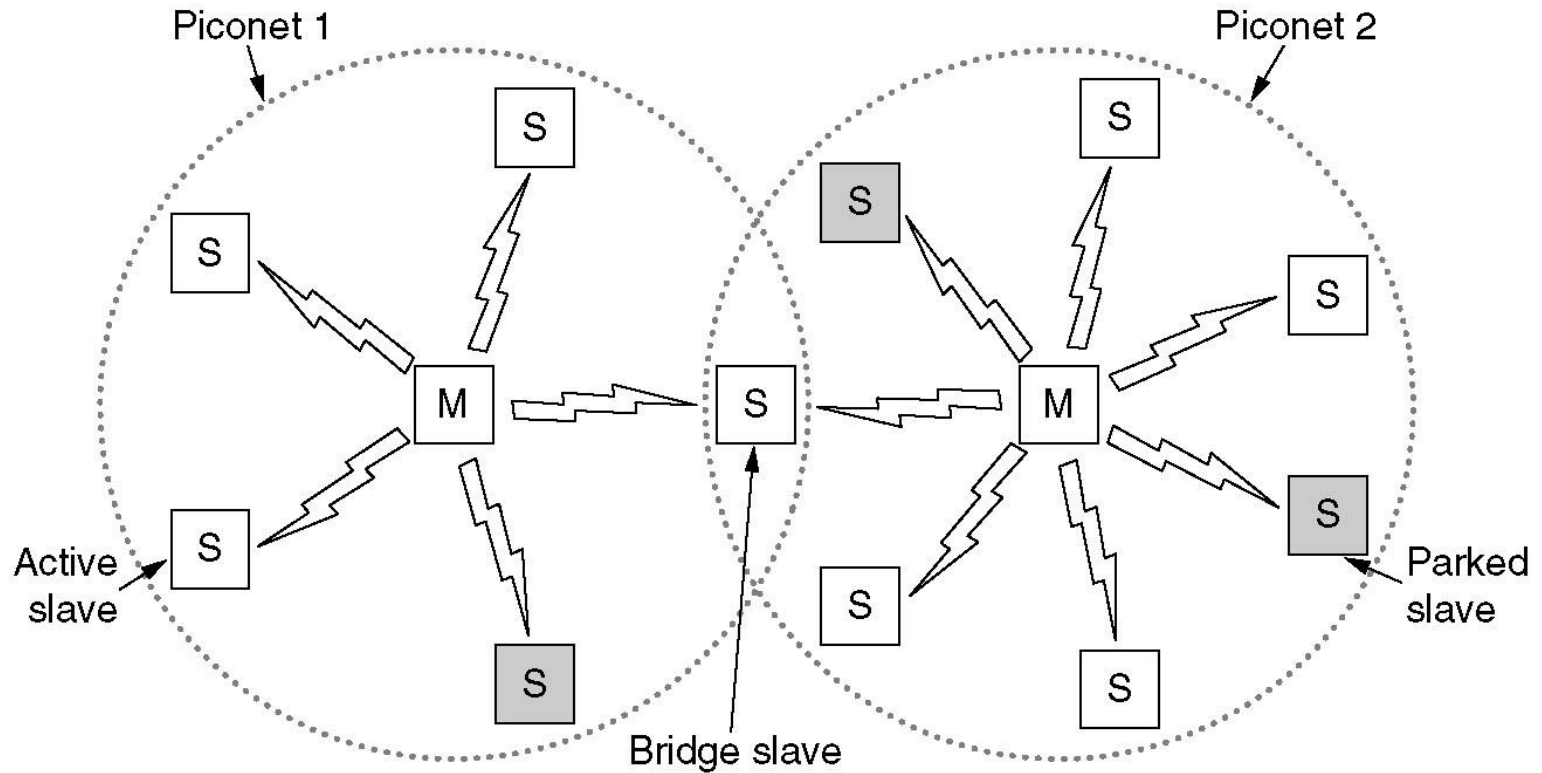


(a) A generic frame. (b) A bandwidth request frame.

Bluetooth

- Bluetooth Architecture
- Bluetooth Applications
- The Bluetooth Protocol Stack
- The Bluetooth Radio Layer
- The Bluetooth Baseband Layer
- The Bluetooth L2CAP Layer
- The Bluetooth Frame Structure

Bluetooth Architecture



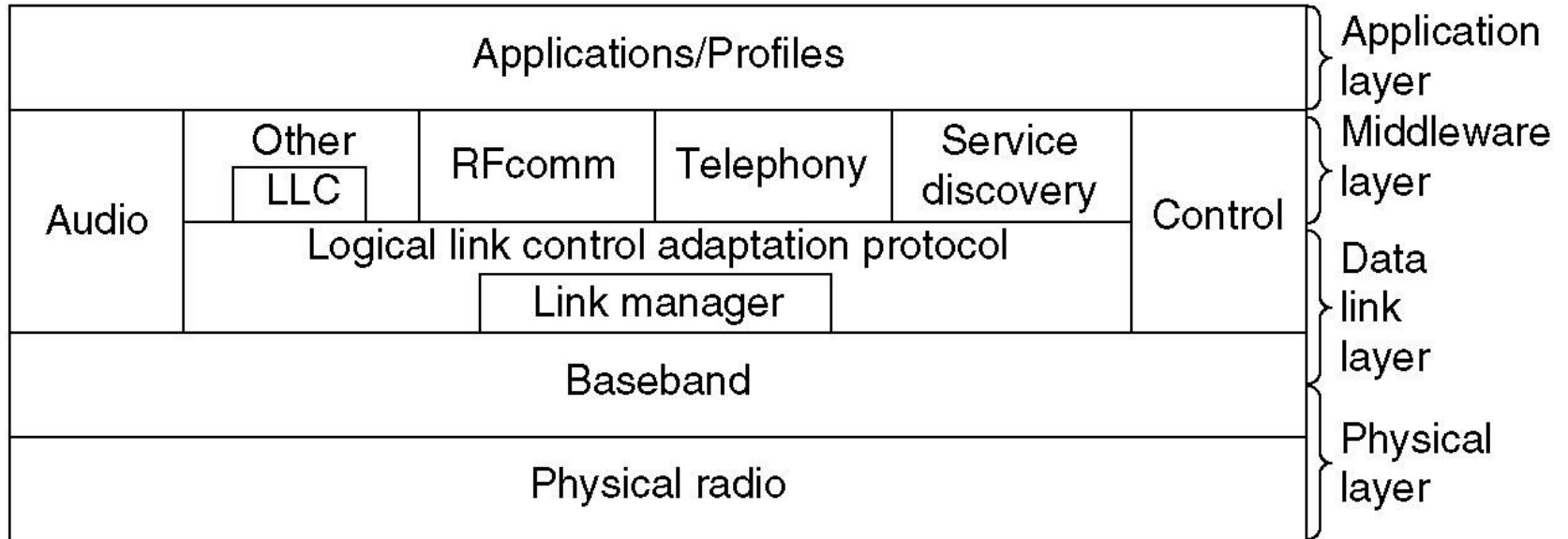
Two piconets can be connected to form a scatternet.

Bluetooth Applications

Name	Description
Generic access	Procedures for link management
Service discovery	Protocol for discovering offered services
Serial port	Replacement for a serial port cable
Generic object exchange	Defines client-server relationship for object movement
LAN access	Protocol between a mobile computer and a fixed LAN
Dial-up networking	Allows a notebook computer to call via a mobile phone
Fax	Allows a mobile fax machine to talk to a mobile phone
Cordless telephony	Connects a handset and its local base station
Intercom	Digital walkie-talkie
Headset	Intended for hands-free voice communication
Object push	Provides a way to exchange simple objects
File transfer	Provides a more general file transfer facility
Synchronization	Permits a PDA to synchronize with another computer

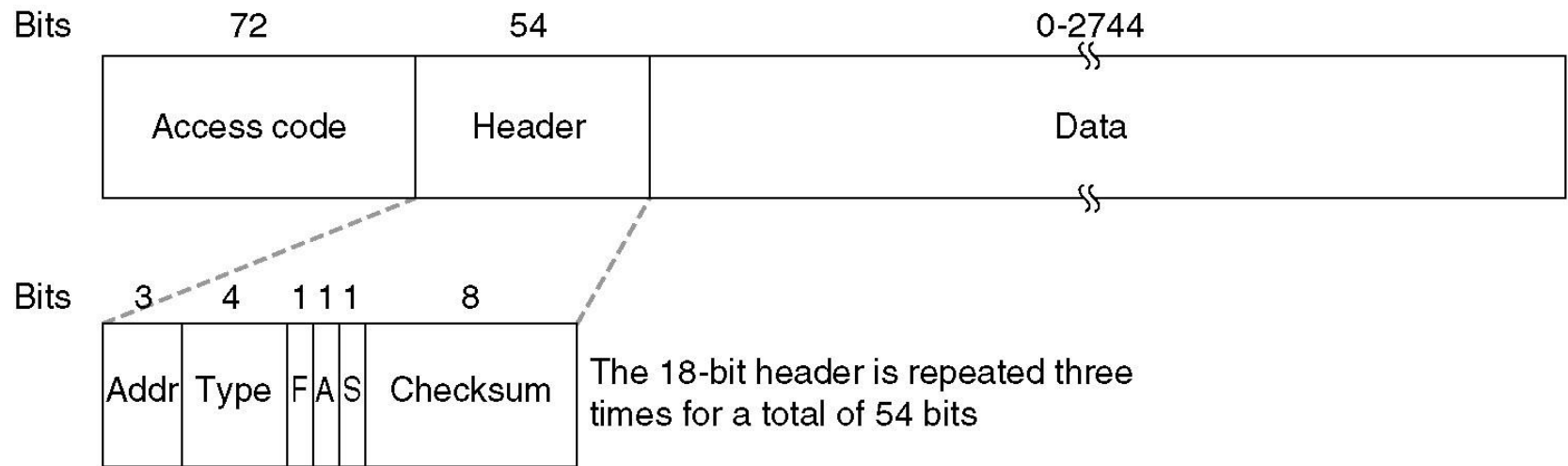
The Bluetooth profiles.

The Bluetooth Protocol Stack



The 802.15 version of the Bluetooth protocol architecture.

The Bluetooth Frame Structure

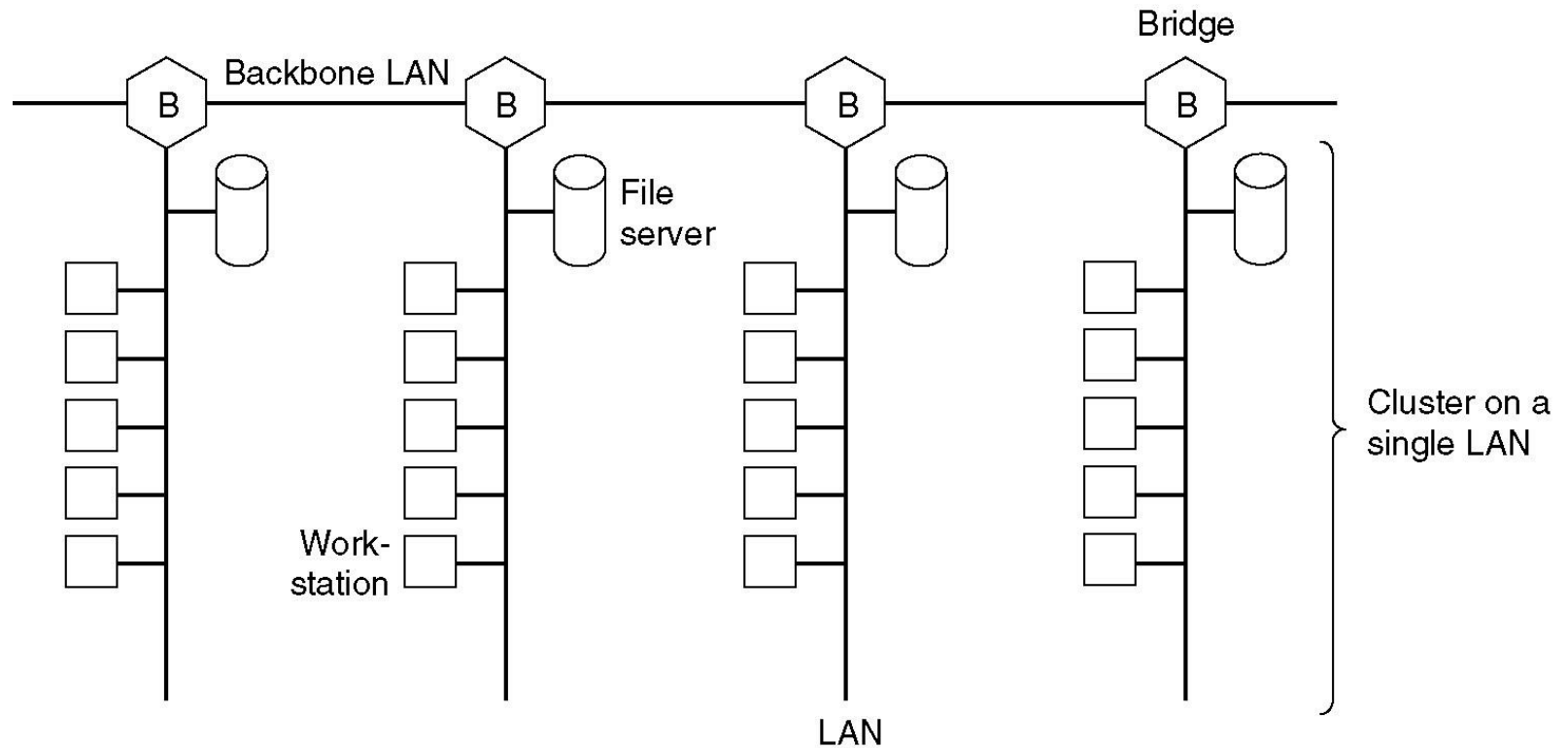


A typical Bluetooth data frame.

6.6 Data Link Layer Switching

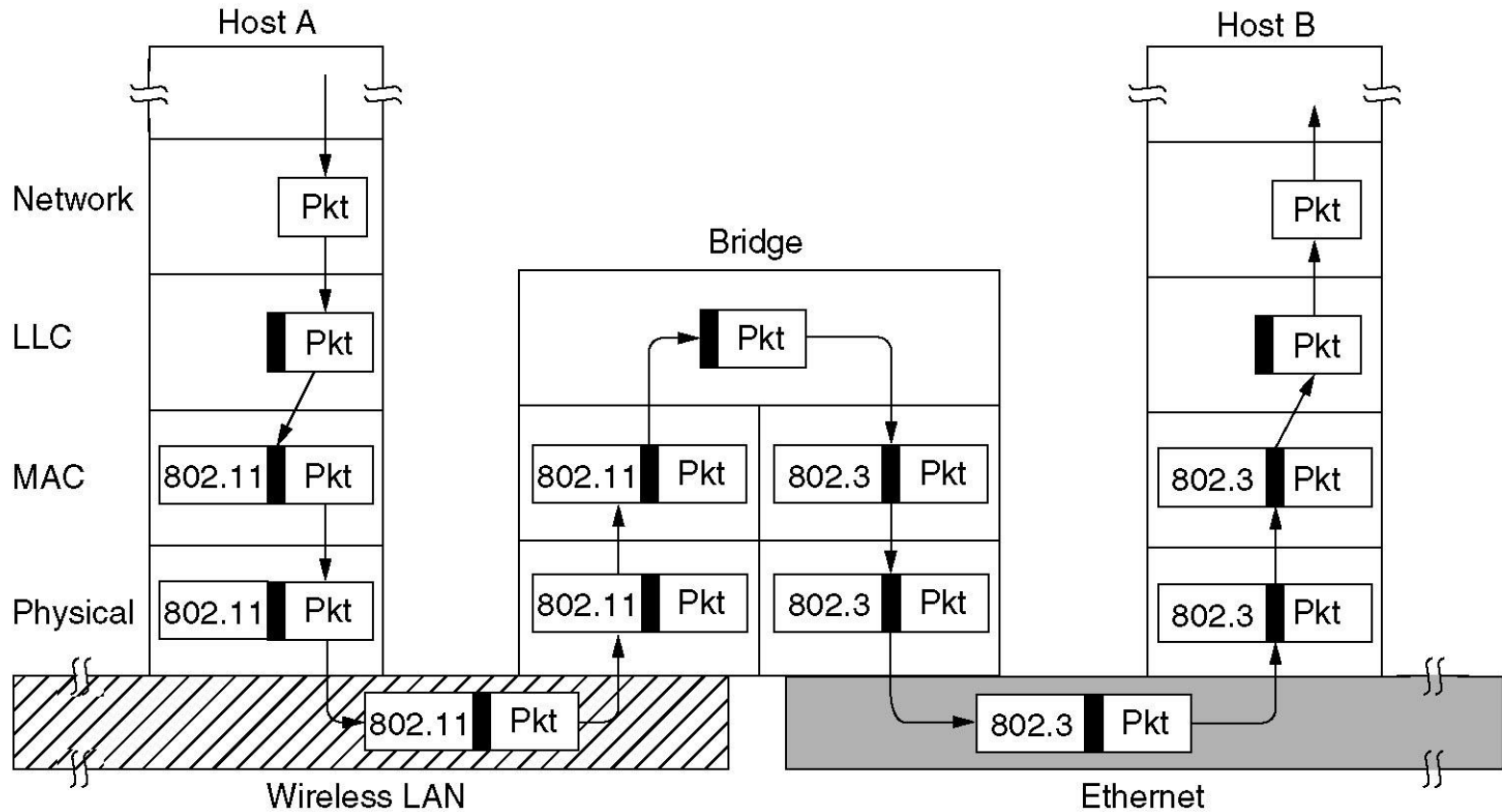
- Bridges from 802.x to 802.y
- Local Internetworking
- Spanning Tree Bridges
- Remote Bridges
- Repeaters, Hubs, Bridges, Switches, Routers, Gateways
- Virtual LANs

Data Link Layer Switching



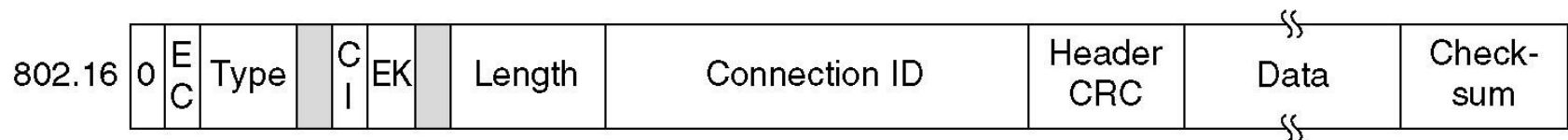
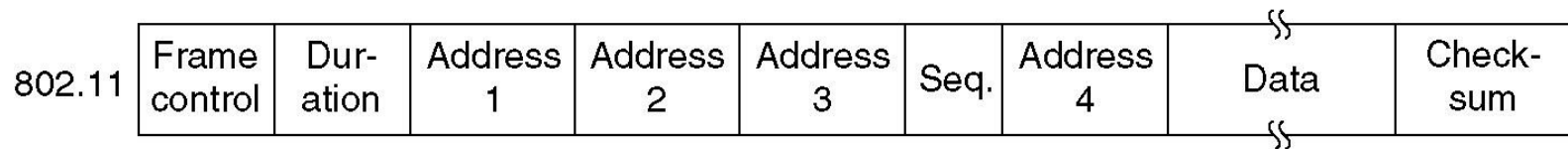
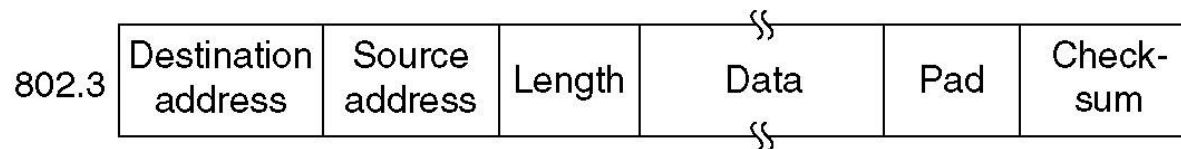
Multiple LANs connected by a backbone to handle a total load higher than the capacity of a single LAN.

Bridges from 802.x to 802.y



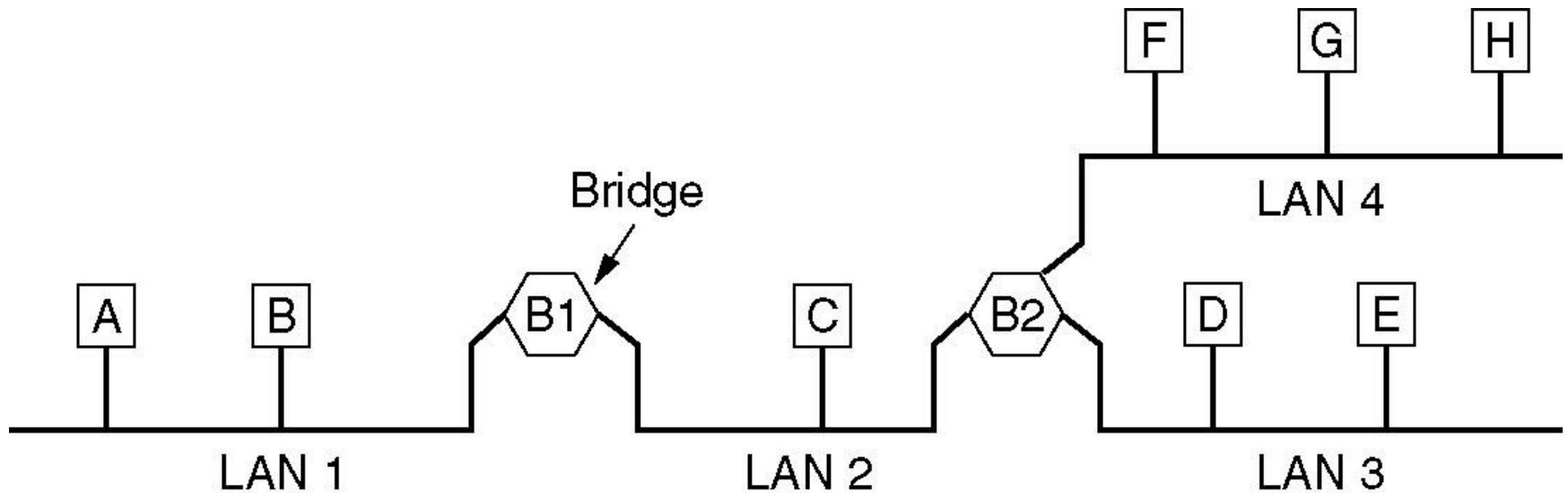
Operation of a LAN bridge from 802.11 to 802.3.

Bridges from 802.x to 802.y (2)



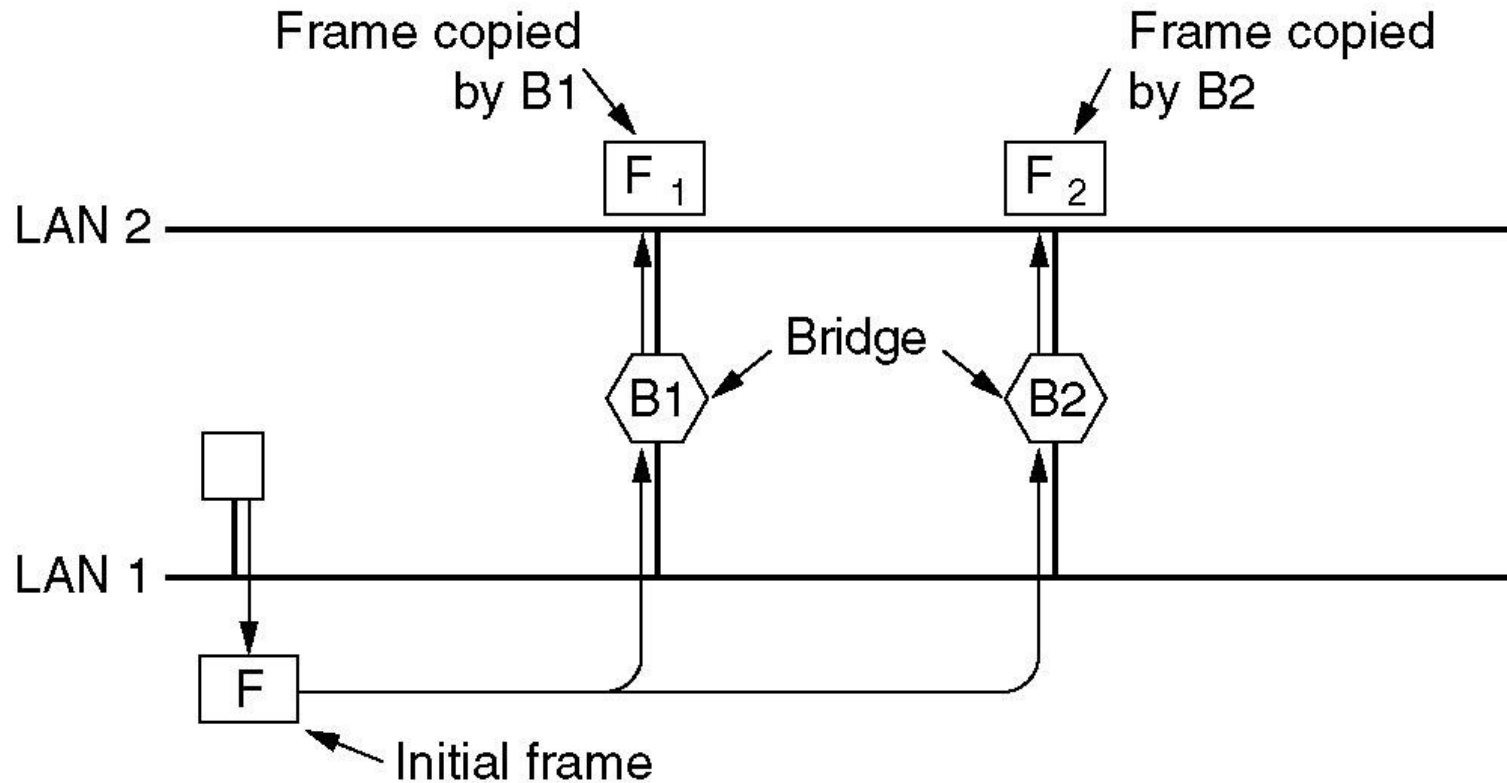
The IEEE 802 frame formats. The drawing is not to scale.

Local Internetworking



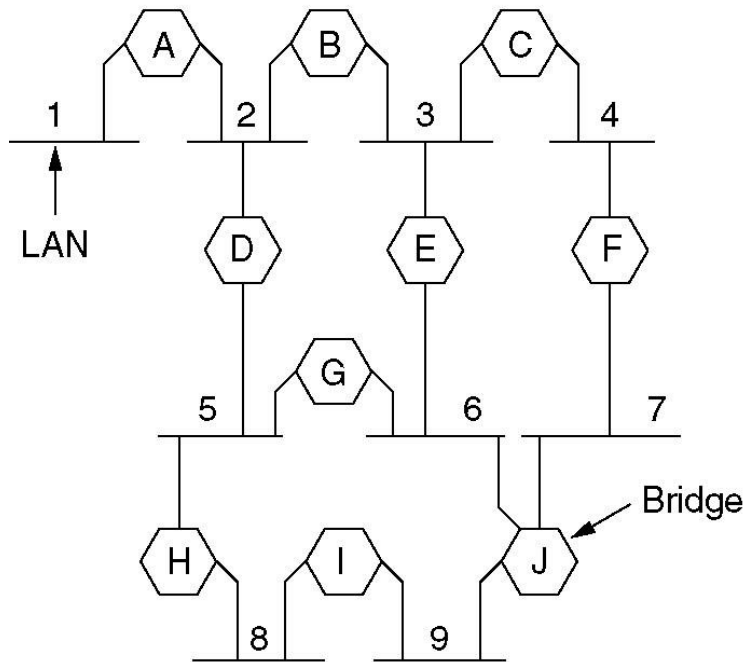
A configuration with four LANs and two bridges.

Spanning Tree Bridges

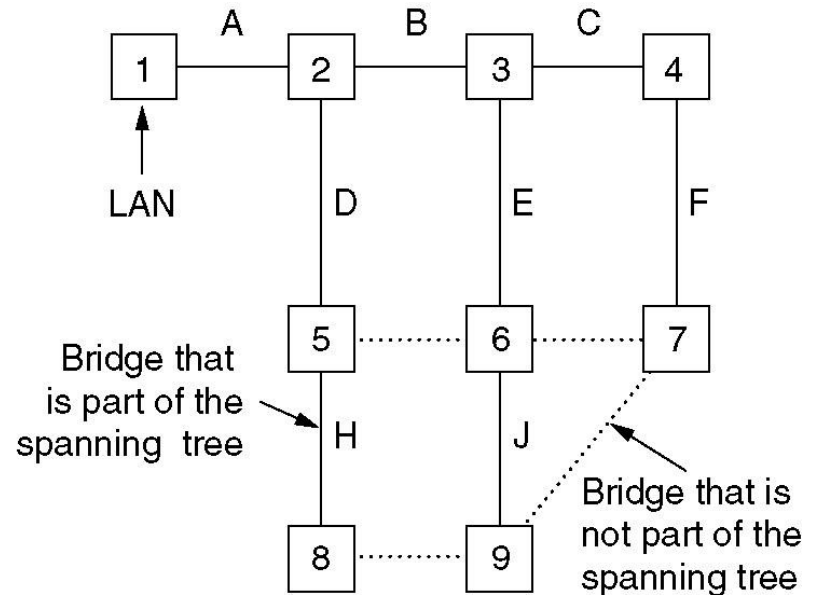


Two parallel transparent bridges.

Spanning Tree Bridges (2)



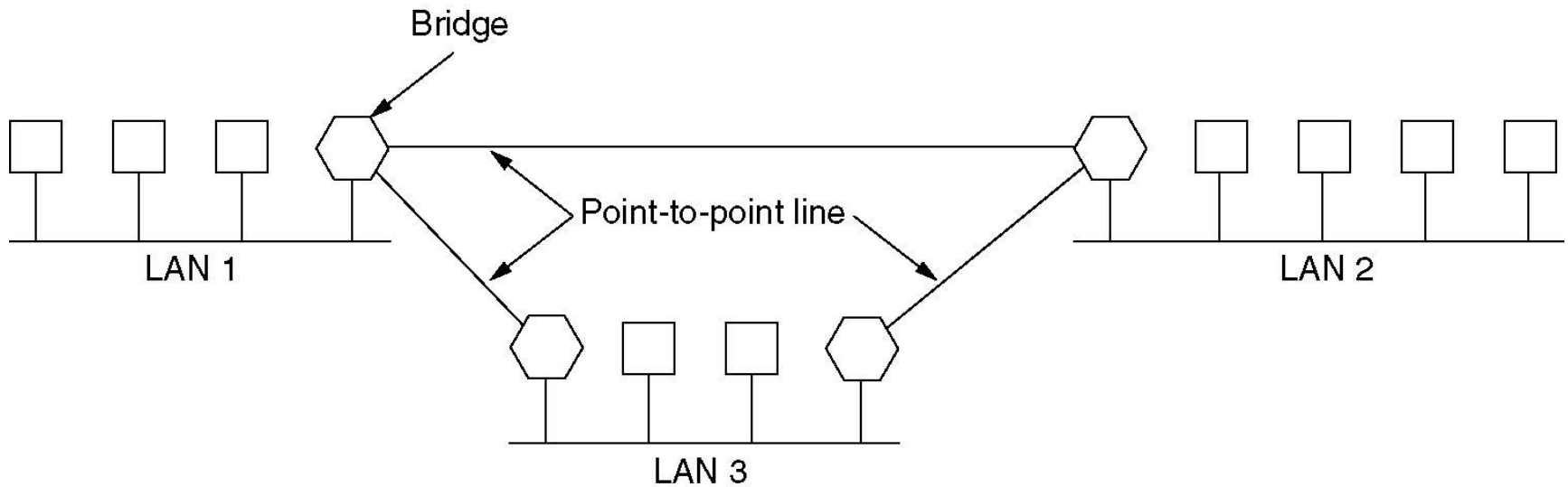
(a)



(b)

(a) Interconnected LANs. (b) A spanning tree covering the LANs. The dotted lines are not part of the spanning tree.

Remote Bridges

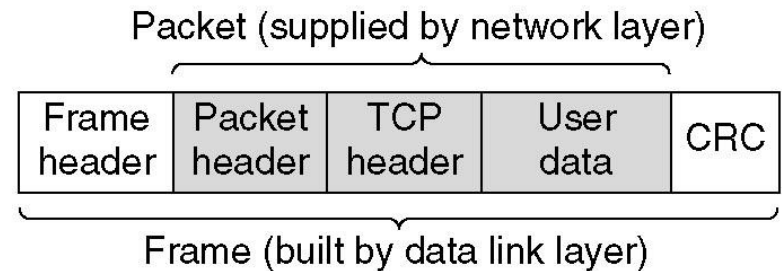


Remote bridges can be used to interconnect distant LANs.

Repeaters, Hubs, Bridges, Switches, Routers and Gateways

Application layer	Application gateway
Transport layer	Transport gateway
Network layer	Router
Data link layer	Bridge, switch
Physical layer	Repeater, hub

(a)

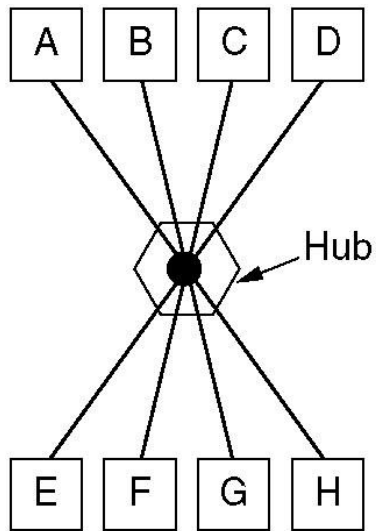


(b)

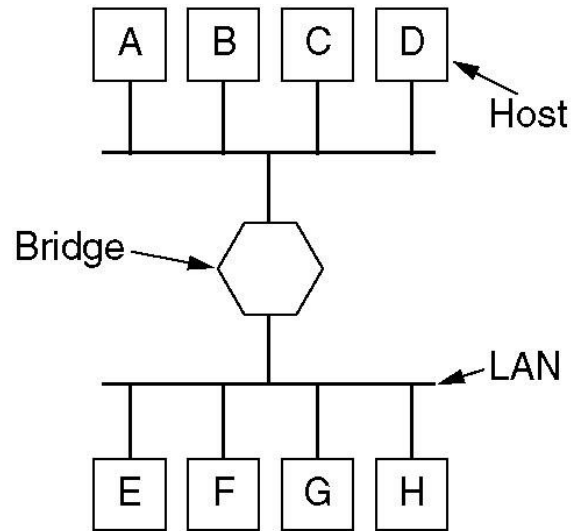
(a) Which device is in which layer.

(b) Frames, packets, and headers.

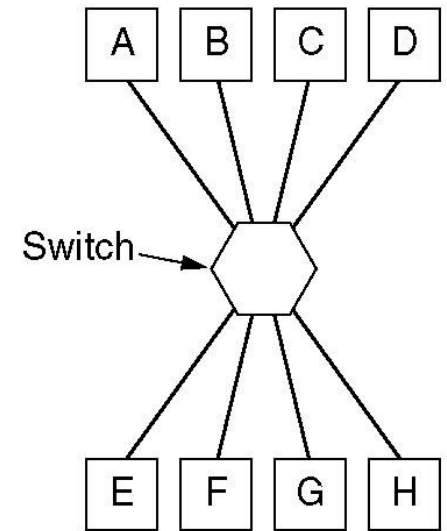
Repeaters, Hubs, Bridges, Switches, Routers and Gateways (2)



(a)



(b)



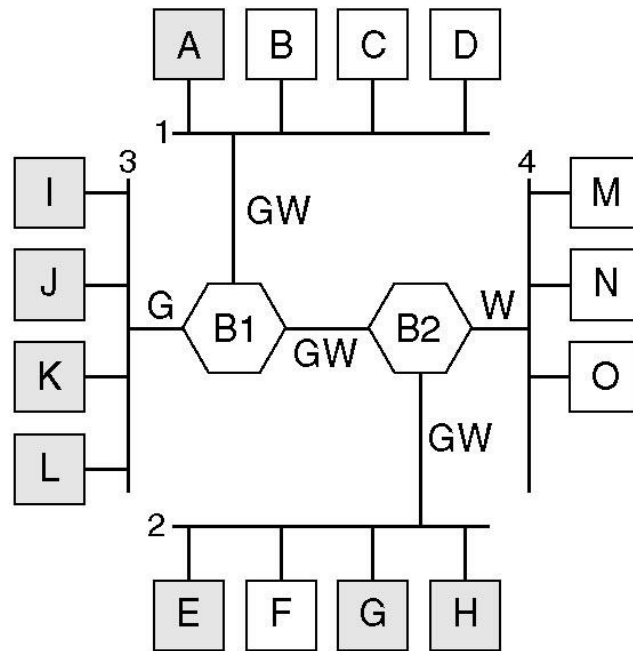
(c)

(a) A hub. (b) A bridge. (c) a switch.

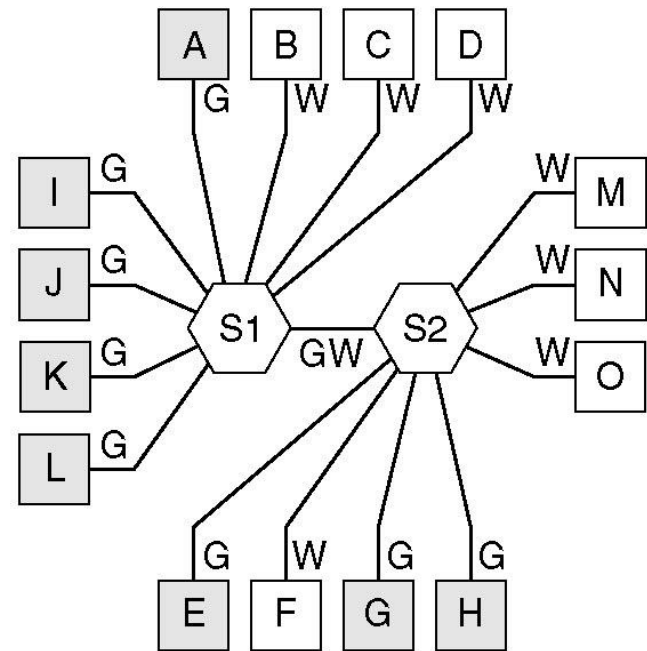
The diagram illustrates a star topology network layout within an office building. It features a central corridor flanked by two rows of offices. Each office contains a computer workstation. On the right side of the corridor, there is a vertical stack of three network devices: a Hub at the top, a Switch in the middle, and another Hub at the bottom. Cables connect each computer workstation to its respective Hub. A thick bundle of cables, labeled 'Twisted pair to a hub', runs along the bottom of the office row. Another thick bundle of cables runs along the top of the office row, connecting to the top Hub. The central corridor is labeled 'Corridor'. The individual offices are labeled 'Office'.

A building with centralized wiring using hubs and a switch.

Virtual LANs (2)



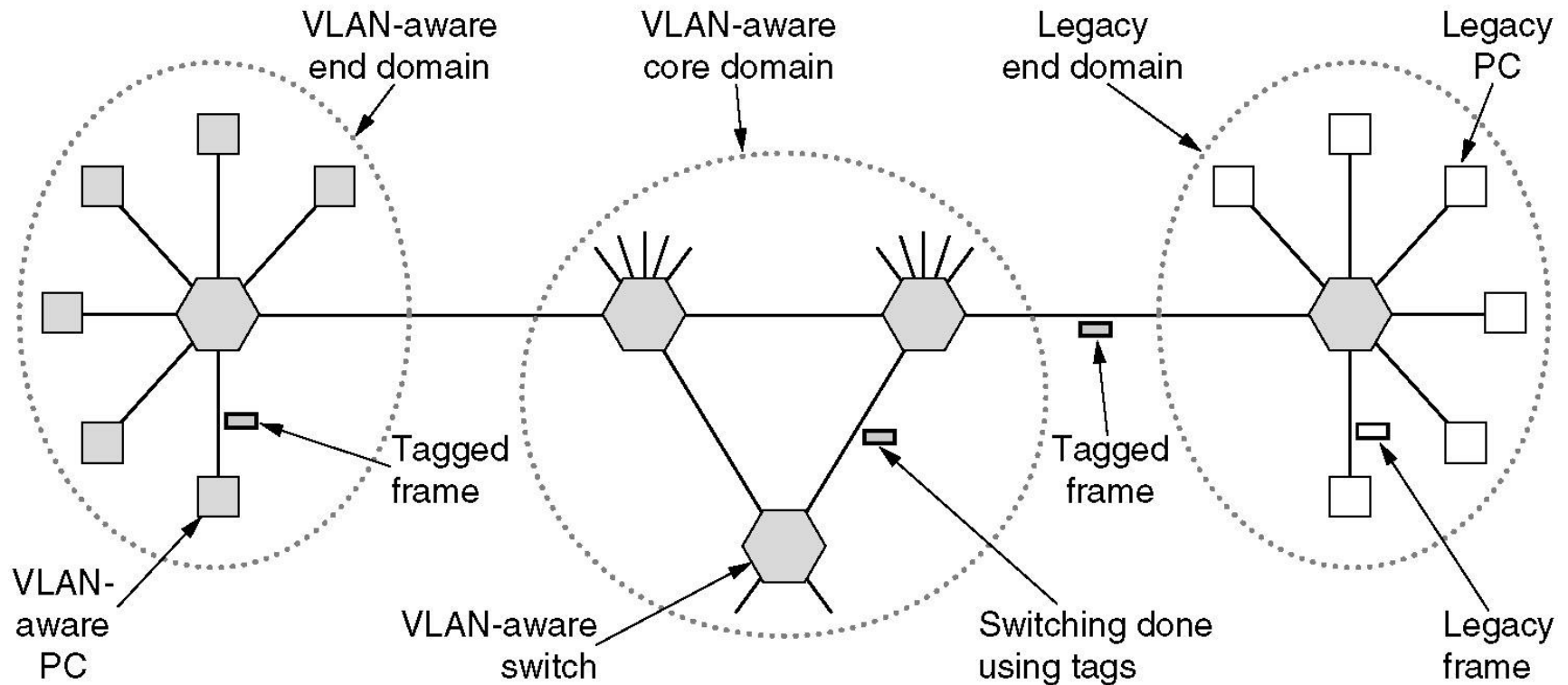
(a)



(b)

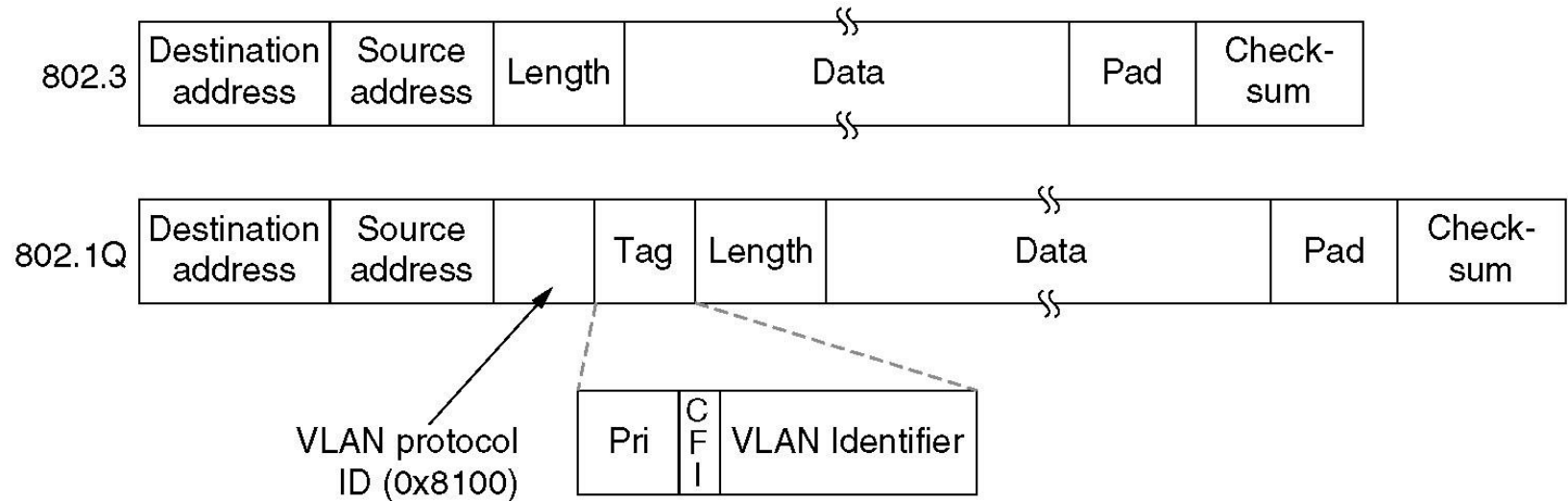
(a) Four physical LANs organized into two VLANs, gray and white, by two bridges. (b) The same 15 machines organized into two VLANs by switches.

The IEEE 802.1Q Standard



Transition from legacy Ethernet to VLAN-aware Ethernet.
The shaded symbols are VLAN aware. The empty ones are not.

The IEEE 802.1Q Standard (2)



The 802.3 (legacy) and 802.1Q Ethernet frame formats.

Summary

Method	Description
FDM	Dedicate a frequency band to each station
WDM	A dynamic FDM scheme for fiber
TDM	Dedicate a time slot to each station
Pure ALOHA	Unsynchronized transmission at any instant
Slotted ALOHA	Random transmission in well-defined time slots
1-persistent CSMA	Standard carrier sense multiple access
Nonpersistent CSMA	Random delay when channel is sensed busy
P-persistent CSMA	CSMA, but with a probability of p of persisting
CSMA/CD	CSMA, but abort on detecting a collision
Bit map	Round robin scheduling using a bit map
Binary countdown	Highest numbered ready station goes next
Tree walk	Reduced contention by selective enabling
MACA, MACAW	Wireless LAN protocols
Ethernet	CSMA/CD with binary exponential backoff
FHSS	Frequency hopping spread spectrum
DSSS	Direct sequence spread spectrum
CSMA/CA	Carrier sense multiple access with collision avoidance

Channel allocation methods and systems for a common channel.